Music and the Piano From the Ground Up

Notes, Scales, Chords, Rhythm and Reading Music, Explained for Complete Beginners

A First-Principles Reader

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Preface

Who this book is for

This book is for adults who have never had a music lesson. You do not need to know what a note, a chord or a beat is. The book starts from the first key you press and explains every word the first time it appears.

By the end you will know your way around a piano keyboard. You will be able to build scales and chords, keep a steady beat, read simple sheet music, and play your first short pieces with both hands.

How to use this book

Music is something you hear, so this book is meant to be heard. In the web edition, every example has a small round play button next to it. Tap it and you hear exactly what the text describes. Try it now with the note called middle C.

The web edition also has keyboards you can tap. When the text says “press this key”, you can press it on the screen and hear it.

A real instrument helps but is not required. A cheap electronic keyboard or a free piano app on your phone is enough. If you have nothing, read with the play buttons and point at the pictures.

Go slowly. Each chapter is short and builds on the one before. When a box says “Try it at the keyboard”, give it a minute before reading on.

A two-minute experiment

Try this now, even before Chapter 1, which explains the names.

1. Find a pair of black keys near the middle of the keyboard. The white key directly to the left of the pair is called C. Put your right thumb on it.
2. Count four keys to the right, counting every key, black and white, but not the one you start on. Put your middle finger there. That key is E.
3. Count three more keys to the right. Put your little finger there. That key is G.
4. Press all three keys together. This is a C major chord. It sounds bright and settled.

5. Keep your thumb and little finger where they are. Move only your middle finger one key to the left, onto a black key. Press the three keys again. This is a C minor chord. It sounds darker and more thoughtful.

Hear the two one after the other. One finger moved by one key, and the whole mood changed. The book works this way throughout: find a pattern on the keys, hear what it does, then give it a name.

Where we are going

- *Chapter 1, The keyboard.* Find every note on the piano, name it, and measure the distance between notes.
- *Chapter 2, Scales and keys.* Build the major scale from one short pattern, and see what "the key of G" means.
- *Chapter 3, Chords.* Build major and minor chords by counting keys, and find the chords that belong together.
- *Chapter 4, Smooth chord changes.* Rearrange chords so your hand barely moves, and accompany a song with two hands.
- *Chapter 5, Rhythm.* Count time: beats, note lengths, rests and time signatures.
- *Chapter 6, Reading notes.* Read notes on the staff, the five lines of sheet music.
- *Chapter 7, Your first pieces.* Put it all together, first a melody, then both hands, with a short daily practice routine.

Why music is in a mathematics curriculum

Music is full of patterns: repeating groups of keys, distances you can count, cycles that come back to where they started. You do not need any mathematics to read this book. Counting up to twelve is enough.

If you do enjoy mathematics, look for the boxes titled "For the mathematically curious". They explain the numbers behind the sound, such as why some notes blend and others clash. They are optional, and in the web edition they start closed. Skip them freely.

Part I

The Keyboard, Scales and Keys

Chapter 1

The Keyboard: Finding Every Note

Press a key at the far left of a piano, then one at the far right. The first growls, the second chirps. In between are 88 keys, black and white, and none of them has a label. How does a pianist find the one they want?

The answer is in the black keys.

1.1 Notes: low and high

Listen to three keys: one near the left end, one in the middle and one near the right end. The first is low, like a big drum. The last is high, like a bird.

The sound one key makes is called a **note**. How high or low a note sounds is called its **pitch**. On a piano, pitch climbs from left to right: every key sounds a little higher than the key to its left.

Hear eight white keys in a row, from left to right. The pitch climbs like feet on a staircase.

Try it at the keyboard

Play a key far to the left, then one far to the right. Then walk up a few white keys from the middle, one at a time, and listen to each note get a little higher.

1.2 The pattern of black keys

Now look instead of listening. The white keys sit side by side with no gaps, so on their own they all look alike. The black keys break the sameness, because they are not spread evenly.

Look at Figure 1.1. The black keys come in a group of two, then a gap, then a group of three, then a gap, then two again, then three, all the way along the keyboard.

Every group of two looks the same, and so does every group of three. That makes them landmarks. A pianist never counts keys from the end of the piano; they look for the nearest group.

One group of two, one group of three and the white keys around them make a block of twelve keys:

seven white and five black. The bracket under the figure marks one block. The whole piano is this block repeated, from the low end to the high end.

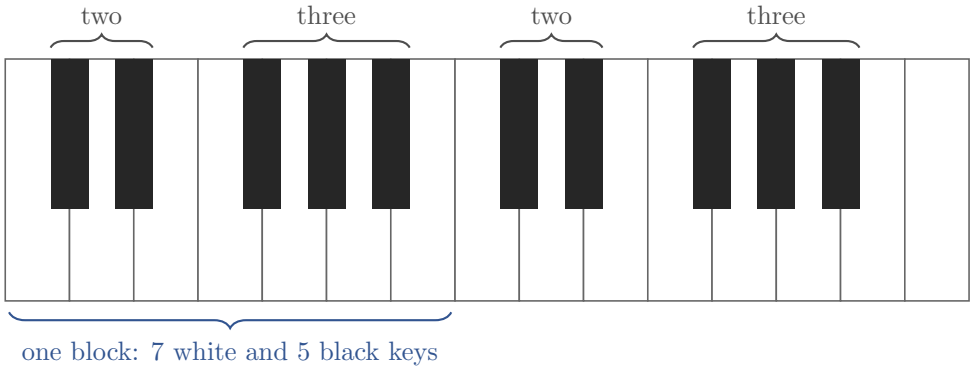


Figure 1.1: The black keys come in groups of two and three. One block of twelve keys, marked below, repeats along the whole keyboard.

Two questions hang over this pattern. Why twos and threes, instead of a black key between every pair of white keys? And why does the block repeat every twelve keys? Your ears will answer both before the end of the chapter.

1.3 Finding C and F

The groups hand you two white keys straight away:

- The white key directly to the left of a group of *two* black keys is called **C**.
- The white key directly to the left of a group of *three* black keys is called **F**.

Every block has a group of two, so every block has a C. Hear three of them, each one block higher than the last. They sound alike, only higher. Keep that in mind; it is a clue. In the same way there is an F before every group of three.

Try it at the keyboard

Find every C on your keyboard, from left to right, by spotting each group of two black keys. Play each one. Then do the same for every F, using the groups of three.

1.4 The seven letter names

C and F are two of the white keys. What are the others called?

Start on C and move to the right. The white keys take the letters in order: C, D, E, F, G, A and B. The next white key is C again, and the letters start over. Music uses only these seven letters, and after G comes A.

Hear one round of letters, from C up to the next C.

Figure 1.2 shows where each letter sits. Use the black keys to find the rest:

- D sits between the two black keys of a group of two. E is directly to the right of that group.
- G and A sit between the black keys of a group of three. B is directly to the right of that group.

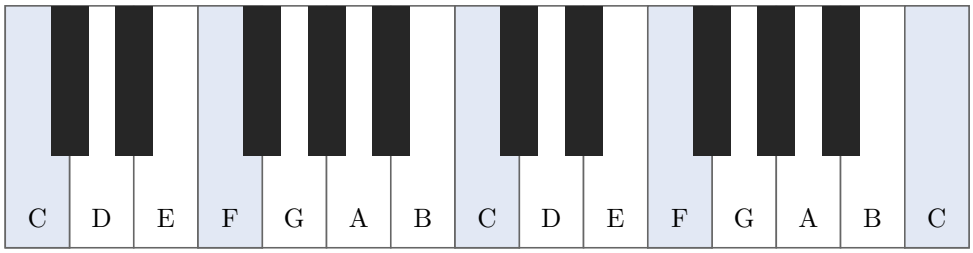


Figure 1.2: The letter names. Every C (shaded) is directly left of a group of two black keys, and every F (shaded) directly left of a group of three. The C nearest the middle of the piano is called middle C.

Try it at the keyboard

Put a finger on any D: it is the white key inside a group of two. Now find any B: directly right of a group of three. Play the white keys from D up to the next D and say each name out loud.

1.5 Octaves and octave numbers

Back to the clue: every C sounds alike. Play a C, then the next C to the right. The second is higher, yet it sounds like the same note sung by a higher voice. Play them together and they blend so well they almost sound like one note.

The distance from a note up to the next note with the same letter is called an **octave**. Going up an octave always means moving twelve keys to the right, counting black and white keys.

This answers the second question. The keyboard repeats every twelve keys because the sound starts over every twelve keys. Each new block holds the same notes as the last one, an octave higher, so it gets the same letters and the same pattern of black keys.

With so many Cs, each needs its own label. The C nearest the middle of the piano is called **middle C**, and it is written C4. The next C up is C5, and the next C down is C3. The number is called the **octave number**. Hear C3, C4 and C5.

An octave number covers every note from a C up to the next B. So the white keys right of middle C are D4, E4, F4 and so on up to B4. Then C5 starts a new octave. The B directly below middle C is B3.

Finger numbers

You now know where middle C is. Before you put a hand on it, one convention. Piano music numbers your fingers so it can say which finger to use. On both hands the thumb is finger 1, the index finger

2, the middle finger 3, the ring finger 4 and the little finger 5.

Try it at the keyboard

Put your right thumb on middle C. Rest fingers 2, 3, 4 and 5 on the next white keys: D, E, F and G. Play them one at a time, fingers 1–2–3–4–5, then back down. This place for your hand is called C position.

1.6 Half steps and whole steps

C, D and E are white keys side by side. So are E, F and G. But between C and D there is a black key, and between E and F there is not. Does that change the sound? Listen to C then D, and then E then F. E to F is a smaller step.

The smallest step on a piano is from one key to the very next key, black or white, with nothing in between. It is called a **half step**. From C to the black key right of it is a half step. From that black key to D is another.

Two half steps make a **whole step**. A whole step skips exactly one key. C to D is a whole step, because it skips the black key between them.

So to measure a distance, count every key, black and white. Each move to the very next key is one half step. Figure 1.3 shows both kinds of step.

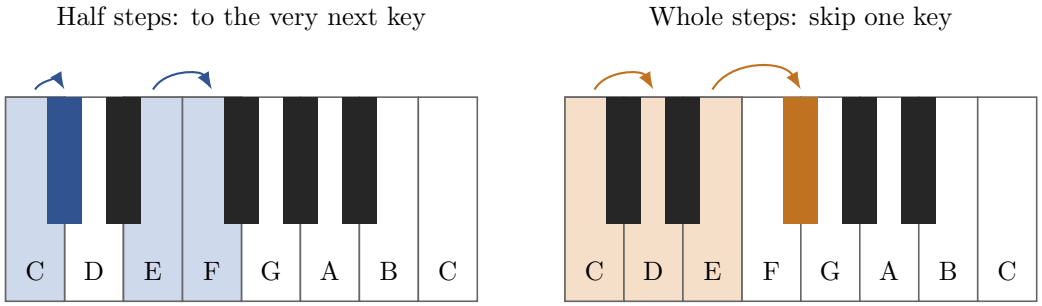


Figure 1.3: Left: C to the black key beside it, and E to F, are both half steps. Right: C to D, and E to the black key directly right of F, are both whole steps.

Look at E and F on the left of the figure. Both keys are white, yet they are only a half step apart, because no black key sits between them. B and C are the same.

Now the first question has an answer. Most neighboring white keys, such as C and D, are a whole step apart, so there is room for one more note between them, and that note is a black key. E–F and B–C are only a half step apart. There is no room between them, so there is no black key. That leaves black keys in two gaps between C, D and E, and in three gaps between F, G, A and B: the twos and threes.

Common trap

Two neighboring white keys are usually a whole step apart, but not always. A whole step up from E is not F: it skips F and lands on the black key directly right of F. In the same way, a

whole step up from B lands on the black key directly right of C.

Try it at the keyboard

Start on C and climb by half steps, touching every key, black and white, up to the next C. You should play 13 keys and take 12 half steps. Then climb from C by whole steps, skipping one key each time.

1.7 Sharps and flats: names for the black keys

The white keys have letters. The black keys have no letters of their own, so they borrow the names of their white neighbors and add a sign:

- A **sharp** (written ♯) means one half step higher, one key to the right. C♯ (said "C sharp") is the key directly right of C.
- A **flat** (written ♭) means one half step lower, one key to the left. D♭ (said "D flat") is the key directly left of D.

The black key between C and D is directly right of C and directly left of D. So it has two names, C♯ and D♭. Both names mean the same key and the same sound. Figure 1.4 shows the two names of every black key.

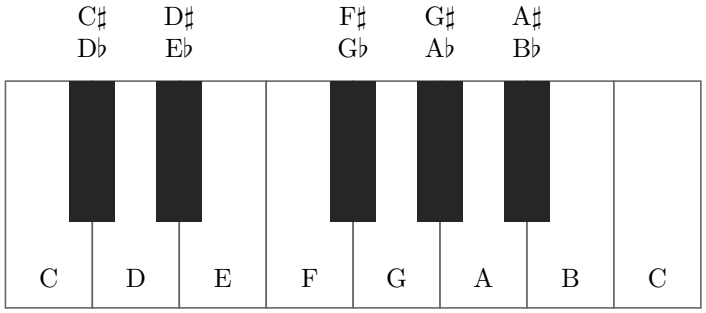


Figure 1.4: Each black key has two names: a sharp of the white key to its left, or a flat of the white key to its right.

Which name do musicians use? It depends on the music around it, and Chapter 2 gives the rule. For now, both are correct.

A sharp or flat does not always land on a black key. E♯ means one key right of E, which is the white key F. C♭ means one key left of C, which is the white key B. These names are rare.

A third sign, the **natural** (written ♮), cancels a sharp or flat. F♮ means plain F, the white key.

Try it at the keyboard

Put a finger on F. Move one key right: that is F♯. Now put a finger on G and move one key left: that is G♭. You are on the same black key both times.

1.8 A first look at intervals

C to E and D to F♯ look different on the keyboard. The first pair is two white keys; the second ends on a black key. Hear them one after the other. The second sounds like the first, moved higher. Why?

Count the keys between them. The distance between two notes is called an **interval**, and you measure it in half steps. Start on the lower note and count one for each key you move to, until you reach the higher note.

From C: C♯ is 1, D is 2, D♯ is 3 and E is 4. From D: D♯ is 1, E is 2, F is 3 and F♯ is 4. Both intervals are 4 half steps, as Figure 1.5 shows. The same count gives the same sound, wherever on the keyboard you start.

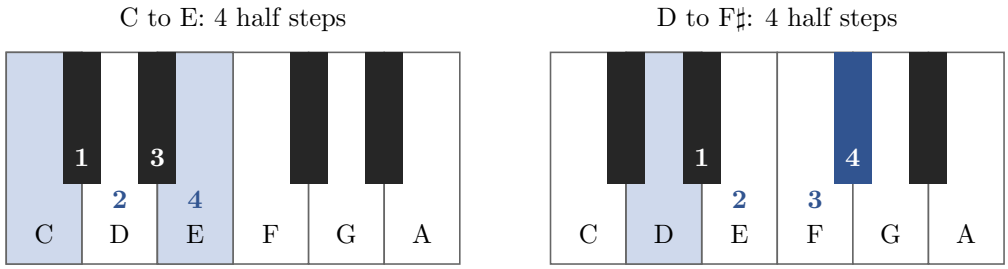


Figure 1.5: Counting half steps. Left: from C, the keys up to E are numbered 1 to 4. Right: from D, the keys up to F♯ are also numbered 1 to 4. Same count, same interval.

The intervals you will hear most often have names. Listen to each one, first as two notes in turn and then together. Chapter 3 builds chords from the two thirds.

- 1 half step, C to C♯: a *half step*. The notes rub against each other.
- 2 half steps, C to D: a *whole step*. A gentler rub.
- 3 half steps, C to E♭: a **minor third**. Soft, a little sad.
- 4 half steps, C to E: a **major third**. Bright and sweet; the notes blend.
- 7 half steps, C to G: a **fifth**. Open and strong, smoother still.
- 12 half steps, C up to the next C: an *octave*. The notes nearly merge.

For the mathematically curious

A note is a vibration. The A above middle C makes the air vibrate 440 times per second; we say its frequency is 440 hertz (Hz).
Going up an octave doubles the frequency: A5 is 880 Hz and A3 is 220 Hz. When one frequency is exactly twice the other, the waves line up every second vibration of the higher note, which is why octaves blend so well.
Other smooth intervals come from other simple ratios. A fifth is close to 3:2 (three vibrations of the top note for every two of the bottom), and a major third is close to 5:4. The half step, close to 16:15, has no simple pattern, and we hear it as a rub.
A modern piano splits each octave into 12 equal steps. Each half step multiplies the frequency by the same number r, and twelve of them make an octave, so $r^{12} = 2$ and $r = 2^{1/12} \approx 1.0595$.

A fifth is then $2^{7/12} \approx 1.498$, very close to 3/2. This system is called equal temperament. Because the names repeat every 12 half steps, the twelve note names behave like the numbers on a clock. Put C at 12 o'clock, C♯ at 1, D at 2, and so on. Counting up 4 half steps from A (at 9 o'clock) lands on C♯ (1 o'clock), in the same way that 4 hours after 9 o'clock is 1 o'clock. Mathematicians call this arithmetic modulo 12.

What to remember

- A *note* is one musical sound; its *pitch* is how high or low it is. Left is lower, right is higher.
- The black keys come in groups of two and three. C is directly left of each group of two; F is directly left of each group of three.
- The white keys are named C, D, E, F, G, A, B, and then the letters repeat.
- An *octave* is 12 keys up, to the next note with the same letter. It sounds like the same note, higher, which is why the keyboard repeats every 12 keys. Middle C is C4; the next C up is C5.
- A *half step* is one key to the very next key. A *whole step* is two half steps. Count every key, black and white. E–F and B–C are half steps, which is why no black key sits between them.
- A *sharp* (♯) is one key higher; a *flat* (♭) is one key lower. Each black key has two names, such as C♯ and D♭.
- An *interval* is the distance between two notes, counted in half steps: a minor third is 3, a major third 4, a fifth 7, an octave 12.

Exercises

Exercise 1.1. Name the white key that sits (a) between the two black keys of a group of two, (b) directly to the right of a group of three, (c) directly to the left of a group of three.

Answer

(a) D. The group of two sits around D: C is on its left, D in the middle, E on its right. (b) B. The group of three has F, G, A and B around it, and B is the white key on its right. (c) F, our second landmark.

Exercise 1.2. Start on B and go (a) one half step up, (b) one whole step up. Then start on F and do the same.

Answer

From B: (a) a half step up is C, a white key, because there is no black key between B and C. (b) A whole step up is two half steps, B to C to C♯, so you land on the black key C♯ (also called D♭). From F: (a) a half step up is the black key F♯. (b) A whole step up skips F♯ and lands on G.

Exercise 1.3. Give both names of (a) the black key between C and D, (b) the middle black key in a group of three. (c) Which white key is F♭?

Answer

(a) C♯ or D♭. (b) The middle black key of the three sits between G and A, so it is G♯ or A♭. (c) F♭ is one key to the left of F. There is no black key there, so it is the white key E.

Exercise 1.4. Count the half steps in each interval, going up: (a) C to G, (b) D to F♯, (c) A to C. What is each interval called?

Answer

(a) C♯ 1, D 2, D♯ 3, E 4, F 5, F♯ 6, G 7: seven half steps, a fifth. (b) D♯ 1, E 2, F 3, F♯ 4: four half steps, a major third. (c) A♯ 1, B 2, C 3: three half steps, a minor third.

Exercise 1.5. (a) What is the C one octave above middle C called? (b) What is the octave number of the B directly below middle C? (c) What is the A directly above middle C called?

Answer

(a) C5. (b) B3: octave numbers change at C, so the B directly below C4 still belongs to octave 3. (c) A4. It is the note that vibrates 440 times per second, used to tune orchestras.

Chapter 2

Scales and Keys

An octave holds twelve different notes, but most songs do not use all twelve equally. They pick seven and stay mostly with those. Which seven? And why does one of them end up sounding like home?

2.1 What a scale is

Play every key from C up to the next C, black and white: twelve half steps in a row. Every step is the same size, so no note stands out. The climb sounds as if it could go on forever.

Now play only the white keys from C up to C. You may know this as do, re, mi, fa, sol, la, ti, do. This time the last C sounds like arriving home.

Stop one note early, on B. It feels unfinished, as if the music is waiting for C.

A set of notes played in order, going up or down like the rungs of a ladder, is called a **scale**. The climb over every key is the **chromatic scale**. The white-key climb is the **C major scale**, and its home note, C, is called its **tonic**.

2.2 The major scale pattern

Why does the white-key climb have a home when the chromatic climb does not? Its steps are not all the same size. Count the half steps between each pair of neighboring white keys from C to C, as in Chapter 1:

- C to D, D to E: whole steps.
- E to F: a half step (no black key between them).
- F to G, G to A, A to B: whole steps.
- B to C: a half step.

Write W for a whole step and H for a half step. The pattern of the major scale is:

W W H W W W H

In Figure 2.1, look at where the two short steps, the H's, fall: between E and F in the middle, and between B and C at the top.

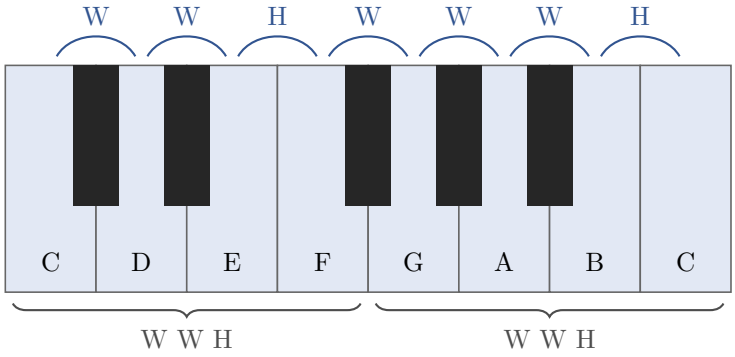


Figure 2.1: The C major scale uses only white keys. Its steps follow the pattern W W H W W W H. The two halves, C to F and G to C, have the same shape.

The half step at the top is the one you heard a moment ago. B is only a half step below C, so close that it leans into C, and C sounds like the place to stop. The chromatic scale has half steps everywhere, so no single note gets that pull. The major scale has only two, and one of them points at home.

The pattern is easier to hold as two halves. Split the eight notes into C–D–E–F and G–A–B–C. Each half has the same shape, W W H, and a whole step, F to G, joins them. Hear the two halves.

Start this pattern on any key and you get a major scale with that key as its tonic. C major is the only major scale on white keys alone, because only from C do the two H's land on E–F and B–C, the two white pairs with no black key between them.

2.3 Building major scales on other notes

G major

Does the pattern still work when it cannot stay on the white keys? Start on G and follow it.

- First half, W W H: G, then a whole step to A, a whole step to B, and a half step to C.
- Cross over with a whole step: C to D.
- Second half, W W H: D, a whole step to E, then a whole step from E. E to F is only a half step, so a whole step from E skips F and lands on the black key F♯. Then a half step from F♯ brings you home to G.

The G major scale is G, A, B, C, D, E, F♯ and G. It has one black key, F♯. Now hear it with a plain F by mistake. F is a whole step below G, too far to lean into it, and the top no longer leads home.

F major

Start on F.

- First half, W W H: F, a whole step to G, a whole step to A, then a half step up from A. One key right of A is the black key we call B♭.
- Cross over with a whole step: B♭ to C.
- Second half, W W H: C, D, E and F, all white keys.

The F major scale is F, G, A, B♭, C, D, E and F. It has one black key, B♭. Figure 2.2 shows G major and F major side by side; look at the one orange black key in each.

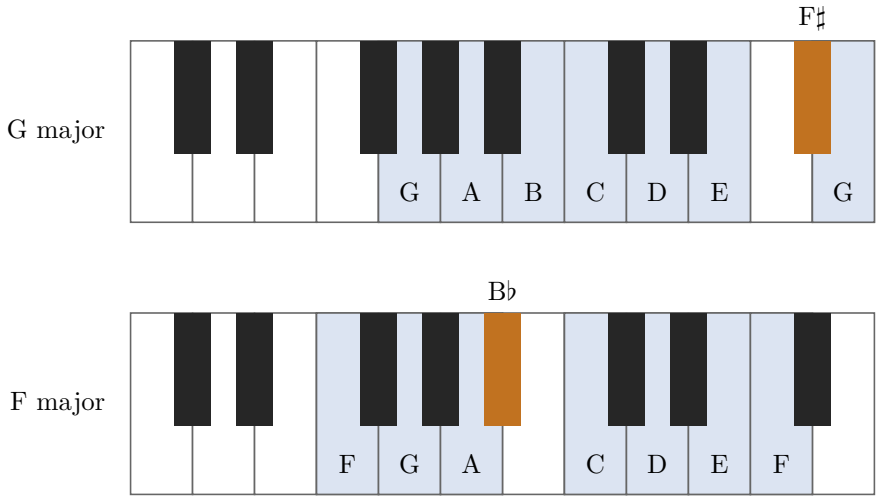


Figure 2.2: G major needs one black key, F♯. F major needs one black key, B♭. All their other notes are white keys.

D major

Start on D: whole step to E, whole step to F♯, half step to G. Cross over: G to A. Then whole step to B, whole step to C♯, half step to D.

The D major scale is D, E, F♯, G, A, B, C♯ and D. It has two black keys: F♯ and C♯.

Try it at the keyboard

Pick a starting key, say A, and build its major scale with the pattern W W H W W W H, counting every key. Check by ear: the last note should feel like home. (Answer: A, B, C♯, D, E, F♯, G♯, A.)

2.4 One letter for each note

Why do we call the black key in F major B♭ and not A♯? They are the same key. The answer is a naming rule.

A major scale uses each of the seven letters exactly once, in order. F major uses F, G, A, B, C, D, E: every letter, none repeated. If we called the fourth note A♯, we would have two kinds of A and no B at all.

The same rule explains F♯ in G major. The scale needs a note between E and G, so it must be some kind of F. That is F♯, not G♭.

So to name the notes of a scale, write the seven letters in order from the tonic first. Then add a sharp or flat to any letter that needs one to fit the pattern.

2.5 What "the key of G" means

Musicians say a song is "in G" or "in the key of G". What does that tell you? Here "key" does not mean a piano key, the thing you press. A musical **key** describes a whole piece of music.

A song in the key of G is built mostly from the notes of the G major scale, and G is its home note. Its melody and chords use G, A, B, C, D, E and F♯. It tends to end on G, because G sounds like home.

So when a musician says "this song is in G", you know two things at once: which seven notes to expect, and which note is home. In the key of G you play F♯ instead of F, almost always.

Each key has its own number of sharps or flats. The key of C has none, G has one sharp (F♯), D has two sharps (F♯ and C♯), and F has one flat (B♭). Sheet music shows these at the start of every line; Chapter 6 shows how this looks on paper.

2.6 Playing the C major scale

You can now build any major scale on paper. How do you play eight notes with five fingers? Pianists tuck the thumb under the hand partway up the scale, so the hand moves along without a jump.

Right hand

Put your right thumb on middle C.

- Going up: play C, D, E with fingers 1–2–3. While finger 3 holds E, pass your thumb under your hand to play F. Then play F, G, A, B, C with fingers 1–2–3–4–5.
- Going down: play C, B, A, G, F with fingers 5–4–3–2–1. Then cross finger 3 over your thumb to play E, and finish E, D, C with 3–2–1.

So the fingers going up are 1–2–3–1–2–3–4–5, and going down they are the same in reverse. Hear it slowly.

Left hand

Put your left little finger on the C below middle C, which is C3.

- Going up: play C, D, E, F, G with fingers 5–4–3–2–1. Then cross finger 3 over your thumb to play A, and finish A, B, C with 3–2–1.

- Going down: play C, B, A with fingers 1–2–3. Pass your thumb under to play G, then play G, F, E, D, C with 1–2–3–4–5.

Hear the left hand going up and down.

G major uses the same fingers in both hands. Start the right hand with the thumb on G4 and the left hand with the little finger on G2, and play F♯ where you played F.

Try it at the keyboard

Play the right hand alone, slowly, until the thumb passes under without a bump. Keep your wrist level and let the thumb slide under a relaxed, rounded hand. Then try the left hand. Speed does not matter yet; evenness does.

2.7 The natural minor scale

The major scale sounds bright. Can the same keys sound darker? Play the white keys again, but start and end on A instead of C: A, B, C, D, E, F, G and A. These are the same seven notes as C major, but now A is home, and the mood turns darker and more serious. A scale with this sound is called a **minor scale**, and this one is the **A natural minor scale** ("natural" here means the plain, basic form; it has nothing to do with the natural sign).

A minor is called the **relative minor** of C major, because the two scales share all their notes. To find the relative minor of any major key, start on the sixth note of its major scale. Or count down 3 half steps from the tonic: from C, down to B, B♭ and A. Figure 2.3 shows the two scales on the same white keys.

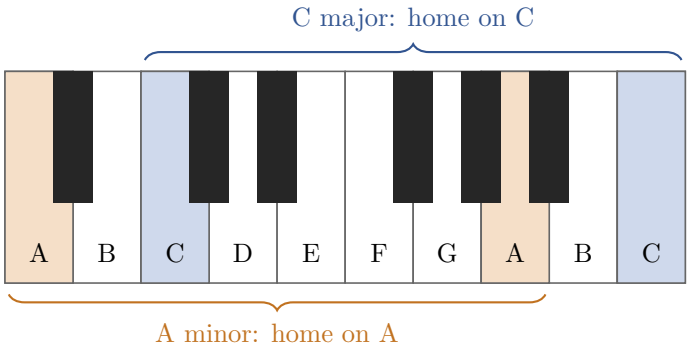


Figure 2.3: C major and A minor use the same seven white keys. Only the home note changes: C for the major scale, A for the minor scale.

Hear C major and then A minor. Same keys, different home, different mood.

Why it sounds darker

Your ear measures every note from the tonic. In C major, the third note, E, is 4 half steps above C: a major third. In A minor, the third note, C, is only 3 half steps above A: a minor third. That smaller distance gives minor its darker color.

Hear C major and C minor, both with C as home. The C minor scale is C, D, E♭, F, G, A♭, B♭ and C. Its notes are the same as E♭ major, because C is 3 half steps below E♭.

Two more minor scales

The relative minor of G major is E minor: E, F♯, G, A, B, C, D and E. The relative minor of F major is D minor: D, E, F, G, A, B♭, C and D.

Listen to the top of A minor. G is a whole step below A, too far to lean into it, so the ending is softer than in major. Composers often raise that note by a half step, to G♯, so it sits right under the tonic the way B sits under C. The result is called the **harmonic minor scale**.

2.8 The pentatonic scale

Half steps give a scale its pull, but two notes a half step apart also rub when they sound together. What happens if you leave them out? Take C major and drop F and B, the top notes of its two half steps. Five notes remain: C, D, E, G and A. A scale of five notes is called a **pentatonic scale**, and this one is the C major pentatonic scale.

With no half steps, none of its notes rub against each other. Almost any order of these five notes sounds like a pleasant tune. That is why pentatonic scales appear in folk songs all over the world.

The five black keys form a pentatonic scale too: G♭, A♭, B♭, D♭ and E♭ (Figure 2.4). Hear them. Started on G♭, they make the G♭ major pentatonic scale.

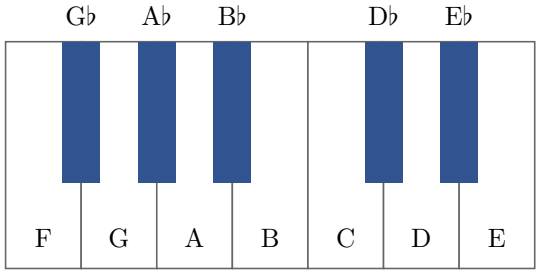


Figure 2.4: The five black keys make a pentatonic scale: G♭, A♭, B♭, D♭ and E♭.

Try it at the keyboard

Improvise on the black keys. With your right hand, play short, slow phrases using only black keys, in any order, with small pauses between them. With your left hand, hold a low G♭ now and then. End a phrase on G♭ and it will sound finished. Here is one idea to start from.

2.9 The circle of fifths

C major has no sharps, G major has one and D major has two. Is there an order to this? Look at the tonics. G is a fifth above C (7 half steps), and D is a fifth above G.

Each time you move up a fifth, the new major scale keeps six notes of the old one and changes one. Which one? The note directly below the new tonic. In C major, F is a whole step below G; G major needs a half step there, so F becomes F♯. From G to D, C becomes C♯, a half step below D, and F♯ stays.

Keep going up by fifths: C, G, D, A, E, B, F♯. Each step adds one more sharp. After twelve steps you are back at C, having visited all twelve keys (past F♯, musicians switch to flat names). Laid out in a circle, as in Figure 2.5, this is the **circle of fifths**.

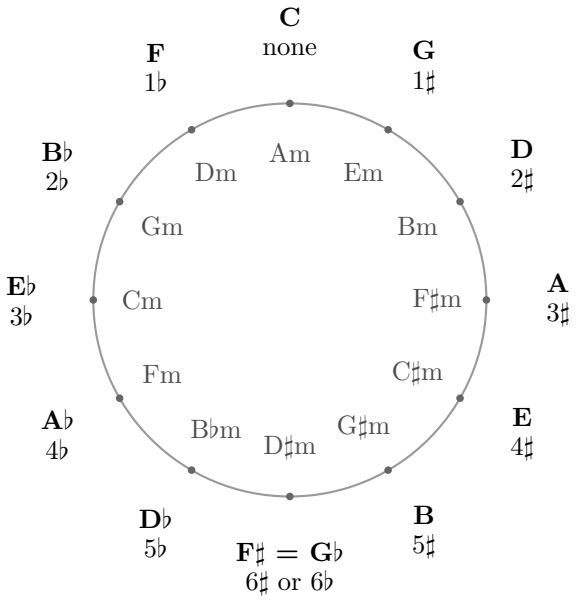


Figure 2.5: The circle of fifths. Going clockwise, each key is a fifth higher and has one more sharp (or one fewer flat). Inside each major key is its relative minor.

Going the other way, counterclockwise, each step moves down a fifth and adds one flat: C, F, B♭, E♭, A♭, D♭, G♭. At the bottom of the circle the two directions meet: F♯ major and G♭ major use exactly the same piano keys, named differently.

A map of neighbors

Two keys next to each other on the circle share six of their seven notes. That is why music often moves to a neighboring key: so little changes that the move sounds like a short trip.

Keys far apart on the circle, such as C and F♯, share few notes. Moving between them sounds far more surprising.

The relative minors sit inside the circle, next to their major partners. A minor is inside C, E minor inside G, and so on. A relative minor has the same sharps or flats as its major partner.

Which sharps, which flats

Sharps always appear in the same order: F♯, C♯, G♯, D♯, A♯, E♯, B♯. A key with two sharps has the first two, F♯ and C♯. A key with three sharps has F♯, C♯ and G♯, and so on.

Flats appear in the reverse order: B♭, E♭, A♭, D♭, G♭, C♭, F♭. A key with three flats has B♭, E♭ and A♭.

Two facts help you check. In a sharp key, the newest sharp is the seventh note of the scale, a half step below the tonic: in D major, C♯ is a half step below D. In a flat key, the newest flat is the fourth note: in F major, B♭ is the fourth note.

For the mathematically curious

Why does stepping by fifths visit all twelve keys before coming back? Number the keys 0 to 11 around the clock (C = 0, C♯ = 1, and so on up to B = 11). A fifth adds 7, and we wrap around after 12, as on a clock.
Starting at 0, the steps give 0, 7, 2, 9, 4, 11, 6, 1, 8, 3, 10, 5, and then 0 again: all twelve numbers. This happens because 7 and 12 have no common factor except 1; in symbols, $\gcd(7, 12) = 1$.
Compare stepping by 3 (a minor third): 0, 3, 6, 9, 0. It returns after only four steps, because 3 divides 12. Stepping by 4 (a major third) visits only three keys. Only steps of 1, 5, 7 and 11 half steps visit every key, and 5 and 7 give the two directions around the circle of fifths.

What to remember

- A *scale* is a set of notes in order. Its home note is the *tonic*.
- The *major scale* follows the pattern W W H W W W H (W = whole step, H = half step): two matching halves, W W H, joined by a W. The last half step, from the seventh note up to the tonic, is what makes the tonic sound like home.
- A major scale uses each letter once. That rule decides whether a black key is called a sharp or a flat.
- A piece in a musical *key*, such as the key of G, uses the notes of that major scale and treats its tonic as home.
- C major scale fingering, right hand going up: 1–2–3, thumb under, 1–2–3–4–5. G major uses the same fingers.
- The *relative minor* starts on the sixth note of a major scale (3 half steps below the tonic) and uses the same notes. Minor sounds darker because its third note is only 3 half steps above the tonic.
- The *pentatonic scale* has five notes and no half steps. The black keys form one.
- On the *circle of fifths*, each step clockwise adds a sharp and each step counterclockwise adds a flat. Neighbors share six of seven notes.

Exercises

Exercise 2.1. Build the E major scale with the pattern W W H W W W H. Which notes are black keys?

Answer

Start on E. Whole step to F♯ (E to F is only a half step, so a whole step skips F), whole step to G♯, half step to A. Whole step to B. Whole step to C♯, whole step to D♯, half step to E. The scale is E, F♯, G♯, A, B, C♯, D♯, E, with four black keys: F♯, G♯, C♯ and D♯.

Exercise 2.2. Build the B♭ major scale, using each letter once. How many flats does it have?

Answer

Start on B♭. Whole step to C (skipping B), whole step to D, half step to E♭. Whole step to F. Whole step to G, whole step to A, half step to B♭. The scale is B♭, C, D, E♭, F, G, A, B♭. It has two flats, B♭ and E♭. We call the fourth note E♭ and not D♯ because the scale already has a D and needs an E.

Exercise 2.3. Find the relative minor of (a) G major, (b) F major, (c) D major. Write out the notes of (c).

Answer

Count down 3 half steps from each tonic, or take the sixth note of the major scale. (a) E minor. (b) D minor. (c) From D, down 3 half steps: C♯, C, B. So B minor. It uses the notes of D major starting on B: B, C♯, D, E, F♯, G, A, B.

Exercise 2.4. Use the order of sharps and flats. (a) A major has three sharps; which ones? (b) E♭ major has three flats; which ones?

Answer

(a) The first three sharps: F♯, C♯ and G♯. Check: G♯ is a half step below A, the tonic. (b) The first three flats: B♭, E♭ and A♭. Check: A♭ is the fourth note of E♭ major (E♭, F, G, A♭).

Exercise 2.5. Which single note is different between (a) C major and G major, (b) C major and F major? Where are these keys on the circle?

Answer

(a) F in C major becomes F♯ in G major. (b) B in C major becomes B♭ in F major. G and F are the two neighbors of C on the circle, one on each side, which is why each differs from C by one note.

Part II

Chords Under Your Hands

Chapter 3

Chords: Notes Played Together

So far you have played notes one after another. A tune you can hum works that way: one note, then the next. A line of notes heard one after another is called a **melody**.

Now press three neighboring keys at the same moment, C, C♯ and D. The notes grind against each other. Then press C, E and G. These three blend into one full, settled sound. Three or more notes played together make a **chord**.

Why do some notes blend and others clash? The answer gives you a recipe for the chords in this chapter.

Chords give a melody its mood. Think of a singer in a room: the melody is the voice, and the chords are the lighting. Hear a short melody alone, then the same melody with chords under it. The sound of chords supporting a melody is called **harmony**.

3.1 Your first chord: C major

Chapter 1 gave you the clue. Two keys side by side, a half step apart, rub against each other. Two keys a third apart (3 or 4 half steps) blend, and a fifth (7 half steps) sounds open and strong.

So build a chord from thirds. Put your right thumb on middle C (C4). Count up 4 half steps: C♯, D, D♯, E. You land on E, a major third above C. From E, count up 3 more: F, F♯, G. You land on G, a minor third above E.

Play C, E and G together. Every pair of notes in it is a friendly interval: C to E is a third, E to G is a third, and C to G, 7 half steps apart, is a fifth. This is the **C major chord**. Its chord symbol is the letter C on its own.

Look at your hand. C, E and G are every other white key: play one, skip one, play one, skip one, play one. Skipping one white key is how you stack one third on top of another. The note a chord is built on, here C, is called its **root**, and it gives the chord its name.

Use fingers 1, 3 and 5: thumb, middle finger and little finger. Fingers 2 and 4 float above the keys you skip. Hear the three notes one at a time, then together.

A chord of three notes built from two stacked thirds is called a **triad** ("tri" means three). Each of

its notes has a name:

- the root is the bottom note, C.
- the **third** is the middle note, E. It is three letters up from the root: C, D, E.
- the **fifth** is the top note, G. It is five letters up: C, D, E, F, G.

Common trap

When you count half steps, count the black keys too. If you count 4 white keys from C, you land on G instead of E.

3.2 Major and minor: one key makes the difference

C major stacks a big third (4 half steps) under a small third (3 half steps). What happens if you swap them?

Keep your thumb on C and your little finger on G. Move only your middle finger one key to the left, from E down to the black key E♭. Play C, E♭ and G together. Now the small third is at the bottom (C up to E♭ is 3) and the big third is on top (E♭ up to G is 4).

This is the **C minor chord**. Its symbol is Cm; the small m means minor. Play C major and then C minor. Most people hear major as bright and settled, and minor as darker or sadder.

Look at Figure 3.1. The outside notes, C and G, stay put, still a fifth apart. Only the middle note moves, and it decides which third sits at the bottom. That gives two recipes that work from any root:

- Major: from the root, count up 4 half steps, then 3 more. We will write this as "4 then 3".
- Minor: count up 3, then 4 more: "3 then 4".

Try the recipes on D. Count 4 up from D: D♯, E, F, F♯. Then 3 more: G, G♯, A. So D major is D, F♯ and A. Move the middle finger down one key to F, and you have D minor, Dm: D, F and A.

Now A. A major is A, C♯ and E, and A minor, Am, is A, C and E.

Why call the middle note of D major F♯ and not G♭, when they are the same key? Because a triad skips one letter between its notes: D, (E), F♯, (G), A. Each note gets the next-but-one letter, the way each scale note in Chapter 2 got the next letter.

Try it at the keyboard

Build G major and G minor, then E major and E minor, by counting. Play each pair and listen for the change of mood. You should find G, B, D and G, B♭, D, then E, G♯, B and E, G, B.

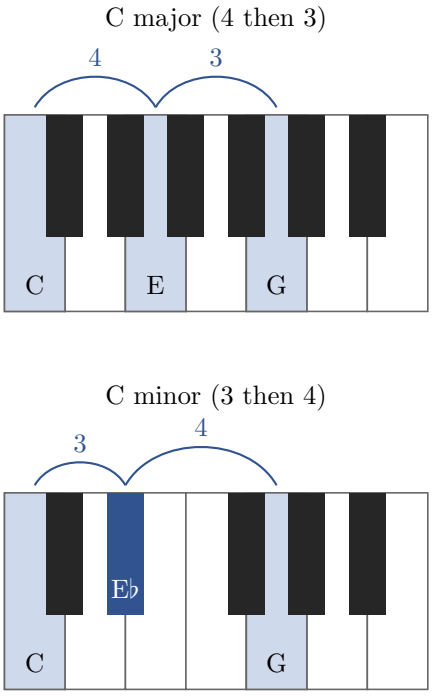


Figure 3.1: C major and C minor share C and G. Only the middle note moves, from E down to E♭, which swaps the big third (4) and the small third (3).

3.3 Three hand shapes for most major chords

Counting always works, but pianists do not count while they play. They remember how each chord looks under the hand. Nine of the twelve major chords fall into three shapes, shown in Figure 3.2:

- White, white, white: C, F and G major.
- White, black, white: D, E and A major. The middle finger sits up on a black key.
- Black, white, black: D♭, E♭ and A♭ major. Thumb and little finger sit on black keys, with the middle finger down on a white key. The notes are D♭, F, A♭; E♭, G, B♭; and A♭, C, E♭.

The other three major chords each have a shape of their own:

- G♭ major (also called F♯ major) is all black keys: G♭, B♭, D♭.
- B♭ major is black, white, white: B♭, D, F.
- B major is white, black, black: B, D♯, F♯.

Intuition

Minor chords swap the first two shapes. Lower the middle note of C, F or G major and you get white, black, white: Cm is C, E♭, G. Lower the middle note of D, E or A major and you get three white keys: Dm, Em and Am.

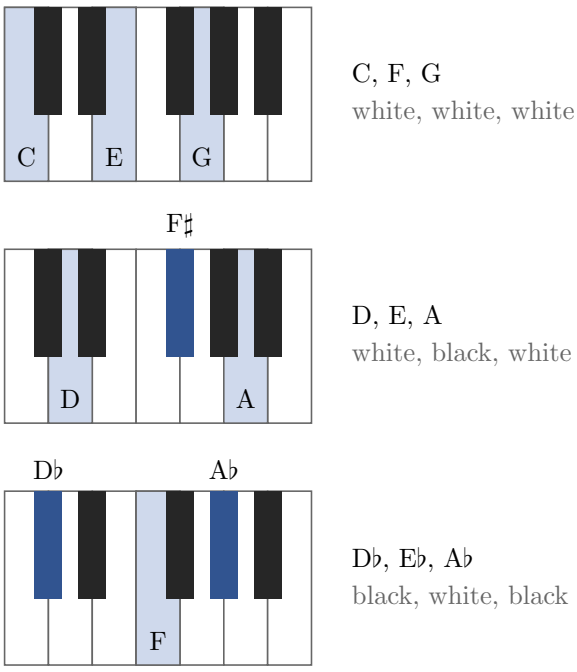


Figure 3.2: Three shapes cover nine major chords. Each keyboard shows one example: C major, D major and D♭ major.

3.4 The chords that belong to a key

A song "in the key of C" (Chapter 2) uses mostly the notes of the C major scale, the white keys, and feels at home on C. Which chords fit such a song? The ones built only from those seven notes.

To stack thirds on white keys only, keep the "play one, skip one" shape. Put fingers 1, 3 and 5 on C, E and G and move your whole hand up one white key at a time. You get seven chords, one on each note of the scale.

Listen again. Your hand kept the same shape, yet some chords sound bright, some dark, and one tense. Why?

Because the white keys are not evenly spaced. There is no black key between E and F, or between B and C, so those steps are half steps. Skipping one white key usually covers 4 half steps, but only 3 when the skip crosses E–F or B–C. Figure 3.3 counts every gap; look at where the 3s appear.

So the shape gives three kinds of chord:

- 4 then 3, major, on C, F and G: C (C, E, G), F (F, A, C) and G (G, B, D).
- 3 then 4, minor, on D, E and A: Dm (D, F, A), Em (E, G, B) and Am (A, C, E).
- 3 then 3 on B: B, D and F. Both of its gaps cross a dashed line.

B, D and F is a **diminished** chord. Its top note is only 6 half steps above the root instead of 7, so it lacks the open fifth, and it sounds tense. It is written Bdim, or B° with a small circle. Hear it lean into C major.

Musicians number the seven chords by the scale note they start on, using Roman numerals: I is 1,

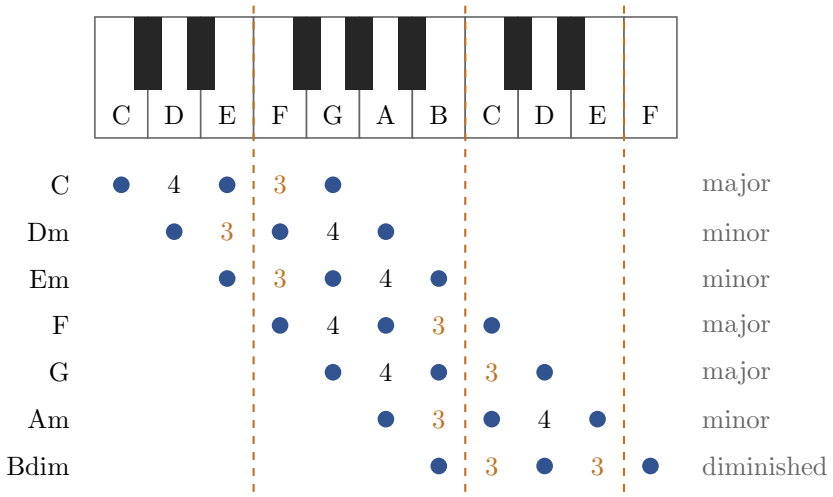


Figure 3.3: Play one, skip one, starting on each white key. The numbers count the half steps between the dots. A gap that crosses a dashed line (E–F or B–C, where there is no black key) is 3; every other gap is 4.

II is 2, III is 3, IV is 4, V is 5, VI is 6 and VII is 7. A capital numeral means a major chord, a small one means minor, and the little circle on vii° means diminished.

Numeral	Built on	Chord	Notes	Sound
I	1st note	C	C, E, G	major
ii	2nd note	Dm	D, F, A	minor
iii	3rd note	Em	E, G, B	minor
IV	4th note	F	F, A, C	major
V	5th note	G	G, B, D	major
vi	6th note	Am	A, C, E	minor
vii°	7th note	Bdim	B, D, F	diminished

Why numbers instead of names? Every major scale has the same pattern of whole and half steps (Chapter 2), so its two half steps always fall in the same places. So every major key gives the same chords: major on 1, 4 and 5, minor on 2, 3 and 6, and diminished on 7.

Check it in the key of G. The G major scale is G, A, B, C, D, E, F♯, and its chords are G, Am, Bm, C, D, Em and F♯dim. In G, the V chord is D. A song described as “I, IV, V” can be played in any key once you know that key’s chords.

Home, going away, and coming home

The chords of a key have different jobs, a bit like places in a story:

- I is home. It sounds settled. Most songs end on it.
- IV and ii step away from home. They give a feeling of setting out.
- V is tension. It pulls strongly back toward I.

Hear the pull of V. In C, play C, F, G and stop: the music feels unfinished. Now let it come home to C.

Why does G pull to C? Remember from Chapter 2 how the C major scale sounded unfinished when it stopped on B. The G chord contains that B, one half step below C. Your ear expects it to rise to C, and when it does, you feel arrival. Figure 3.4 shows the move.

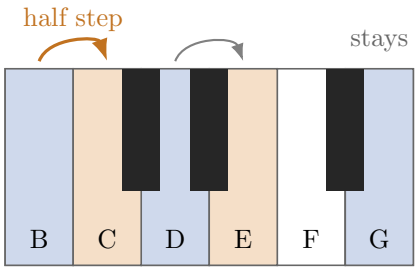


Figure 3.4: From G (blue: B, D, G) to C (C, E, G). G stays, D steps up to E, and B rises only a half step to C. That smallest step is the strongest pull home.

The vi chord, Am, is a quieter relative of home: it shares two notes, C and E, with the C chord.

3.5 Famous chord progressions

A **chord progression** is a chain of chords that a song repeats, usually with a steady number of counts on each chord. A few progressions turn up in thousands of songs. Here are four, in the key of C. Each button plays the chords once, two counts each, then lands on the first chord again.

- I–V–vi–IV: C, G, Am, F. One of the most common progressions in pop music.
- vi–IV–I–V: Am, F, C, G. The same four chords, starting on the minor one, so it feels more wistful.
- I–vi–IV–V: C, Am, F, G. The sound of many 1950s songs.
- ii–V–I: Dm, G, C. A short trip home, used everywhere in jazz.

To move a progression to another key, keep the numerals and read the chords from the new key. In the key of G, I is G, ii is Am, IV is C, V is D and vi is Em:

Numerals	In C	In G
I–V–vi–IV	C, G, Am, F	G, D, Em, C
vi–IV–I–V	Am, F, C, G	Em, C, G, D
I–vi–IV–V	C, Am, F, G	G, Em, C, D
ii–V–I	Dm, G, C	Am, D, G

Hear I–V–vi–IV in G and ii–V–I in G.

Try it at the keyboard

Play I–V–vi–IV in C with your right hand: C, G, Am, F. Count 1, 2, 3, 4 slowly and steadily, and change chord each time you get back to 1. Your hand jumps around for now. Chapter 4 shows how to make these changes with almost no movement.

3.6 Sus chords: leaving out the third

The middle note of a triad decides major or minor. What if there is no third at all?

Play C major, then lift your middle finger off E and put finger 4 on F. You now hold C, F and G. This is **Csus4** (said "C sus four"). "Sus" is short for suspended: the middle note hangs one key above the third, so the chord is neither major nor minor.

A sus chord sounds open and unfinished. It wants the F to fall back to E. Hear the pull from Csus4 to C. With finger 2 on D instead, C, D and G is **Csus2**.

Here are all the triad recipes in this chapter, with C as the root:

Chord	Symbol	Count up	Notes from C
Major	C	4 then 3	C, E, G
Minor	Cm	3 then 4	C, E♭, G
Diminished	Cdim	3 then 3	C, E♭, G♭
Sus4	Csus4	5 then 2	C, F, G
Sus2	Csus2	2 then 5	C, D, G

3.7 Seventh chords: one more third

A triad is two thirds stacked. Stack one more: play one, skip one, four times. From C you get C, E, G and then B. A four-note chord like this is a **seventh chord**, because its top note is seven letters above the root: C, D, E, F, G, A, B.

Three seventh chords are worth knowing first. Their symbols tell them apart:

- Cmaj7 ("C major seven") is C, E, G, B. It sounds soft and dreamy.
- Dm7 ("D minor seven") is D, F, A, C. It sounds mellow.
- G7 ("G seven") is G, B, D, F. It pulls toward C even harder than G does. B still wants to rise a half step to C, and now F also wants to fall a half step to E.

Common trap

C7 and Cmaj7 are different chords. C7 is C, E, G, B♭: its top note is the black key B♭. Cmaj7 is C, E, G, B. A plain 7 always means the lower top note, which sounds bluesy and restless; maj7 means the higher one. Compare them.

Four notes in one hand can be a stretch. Instead, play the root low with your left hand and the other three notes with your right. Those top three notes always form a triad you already know:

- Cmaj7 is C under E, G, B. The right hand plays Em.
- Dm7 is D under F, A, C. The right hand plays F major.
- G7 is G under B, D, F. The right hand plays Bdim.
- Am7 (A, C, E, G) is A under C, E, G. The right hand plays C major.

Work it yourself

Play ii–V–I in C with seventh chords: Dm7, G7, Cmaj7. Count 4 slow counts on each chord. Left hand: D3, then G2, then C3. Right hand, above middle C: F, A, C, then F, G, B, then E, G, B. (In G7 the right hand leaves out D; you will not miss it.) Your right hand barely moves.

For the mathematically curious

Why do thirds and fifths blend? Each note is a vibration (Chapter 1). In a major chord the three notes vibrate at speeds close to the ratio 4:5:6, so their waves line up again and again. Neighboring keys, close to 16:15, almost never line up, and we hear the clash.

Place the twelve notes around a clock face, C at 12 o'clock, one note per hour: C♯ at 1, D at 2, and so on. Join the three notes of a major chord and you get a triangle whose sides span 4, 3 and 5 hours. Every major chord is the same triangle, turned to a new position. That is why one counting recipe works from every root.

A minor chord is the same triangle flipped over: 3, 4 and 5 hours. A diminished chord is 3, 3 and 6. The chord C, E, G♯ (4, 4 and 4) makes an equilateral triangle; it is called augmented.

What to remember

- A *melody* is notes one after another; a *chord* is notes played together; *harmony* is the sound of chords under a melody.
- A *triad* stacks two thirds: the *root* (which names it), the *third* and the *fifth*. On the white keys it is "play one, skip one". Play it with fingers 1, 3 and 5.
- Major: count up 4 half steps, then 3 more. Minor: 3, then 4. Only the middle note differs, by one key.
- The chords of a major key are always I, ii, iii, IV, V, vi, vii°. In C: C, Dm, Em, F, G, Am, Bdim. The pattern comes from where the scale's two half steps fall.
- A *diminished* chord counts 3 then 3 and sounds tense. A *sus* chord replaces the third and sounds open and unfinished.
- I is home, IV steps away, and V pulls back home because it holds the note a half step below the root of I. A *chord progression* is a chain of chords a song repeats, such as I–V–vi–IV.
- A *seventh chord* stacks one more third. G7 and Gmaj7 are different chords. Play the root in the left hand and a familiar triad in the right.

Exercises

Exercise 3.1. Build E major and E♭ major by counting 4 then 3. Which of the three shapes does each belong to? Then lower the middle note of each to get Em and E♭m.

Answer

From E, count 4 half steps: F, F♯, G, G♯. Then 3 more: A, A♯, B. So E major is E, G♯, B. That is white, black, white.
From E♭, count 4: E, F, G♭, G. Then 3 more: A♭, A, B♭. So E♭ major is E♭, G, B♭. That is black, white, black.
Lowering the middle notes gives Em: E, G, B, all white keys. And E♭m: E♭, G♭, B♭, all black keys. We call the middle note G♭ rather than F♯ so that the letters skip one each time: E, G, B.

Exercise 3.2. Spell Dsus4 (5 then 2 from D) and Dsus2 (2 then 5 from D). Compare them with D major.

Answer

Dsus4: from D, count 5 half steps to G, then 2 more to A. So Dsus4 is D, G, A. Dsus2: count 2 to E, then 5 more to A. So Dsus2 is D, E, A.
D major is D, F♯, A. All three share D and A. Only the middle note changes: G for sus4, F♯ for major, E for sus2. Hear Dsus4 settle into D major.

Exercise 3.3. The F major scale is F, G, A, B♭, C, D, E. List the seven chords of the key of F, with their Roman numerals. Then play I–V–vi–IV in F.

Answer

Use the "play one, skip one" shape on the notes of the F major scale. You get: I is F (F, A, C); ii is Gm (G, B♭, D); iii is Am (A, C, E); IV is B♭ (B♭, D, F); V is C (C, E, G); vi is Dm (D, F, A); vii° is Edim (E, G, B♭).
So I–V–vi–IV in F is F, C, Dm, B♭.

Exercise 3.4. A song in the key of G uses only three chords: G, C and D. What are their Roman numerals? Which three chords would the same song use in the key of D? (The D major scale is D, E, F♯, G, A, B, C♯.)

Answer

In G, the chord on the 1st note is G, the 4th note is C, and the 5th note is D. So the song uses I, IV and V. In the key of D, the 1st, 4th and 5th notes are D, G and A, so the song uses D, G and A major.

Exercise 3.5. Em7 is E, G, B, D. Which triad does the right hand play over a left-hand E? And D7 is D, F♯, A, C: which two of its notes move by a half step to reach the G chord (G, B, D)?

Answer

Take away the root E and you are left with G, B, D: a G major triad. So Em7 is E in the left hand under G major in the right hand.
In D7, F♯ rises a half step to G, and C falls a half step to B. That is the same pull you met in G7 going to C, moved to a new key.

Chapter 4

Inversions and Smooth Chord Changes

Play C, F, G and back to C with the chord shapes from Chapter 3. Your whole hand jumps up to F, then higher to G, then all the way back down. Each jump takes time, and the music stumbles while your hand is in the air.

An experienced pianist plays the same three chords with a hand that hardly moves. The chords are the same, but the second version flows. How can the same chords need less movement? The pianist rearranges the notes inside each chord.

4.1 Same notes, different bottom

Can you rearrange a chord and keep it the same chord? Chapter 1 says yes. Notes an octave apart sound like the same note, higher or lower. So if you move one note of C major up an octave, you still hear C, E and G.

Start with C4, E4, G4. This is **root position**: the root is at the bottom. Now take the bottom note, C4, and move it up to the next C, C5. You get E4, G4, C5: the **first inversion**. Do it again: move E4 up to E5, and you get G4, C5, E5, the **second inversion**. Hear all three.

All three sound like C major. The top note changes, so each has a slightly different color, but none of them sounds like a new chord. A chord played with a note other than the root at the bottom is called an **inversion**. Figure 4.1 shows the three positions; follow the darker key, C, as it climbs.

Each position has its own fingering:

- Root position, C4–E4–G4: fingers 1–3–5.
- First inversion, E4–G4–C5: fingers 1–2–5. The index finger takes G, which leaves room for the bigger stretch from G up to C.
- Second inversion, G4–C5–E5: fingers 1–3–5. Here the big stretch is at the bottom, between thumb and middle finger.

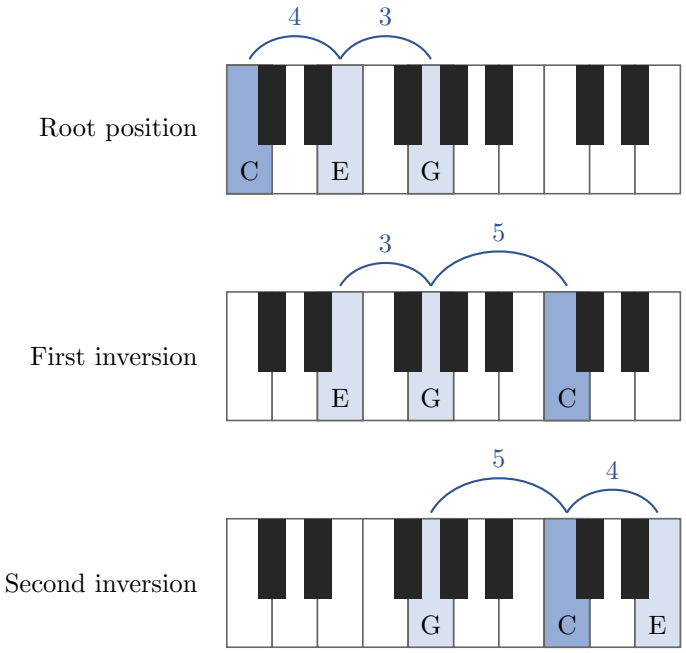


Figure 4.1: The three positions of C major. The root, C, is the darker key. The numbers count the half steps between neighboring notes; in both inversions the 5 sits directly below C.

Try it at the keyboard

Play the three positions of C major one after another, going up: C4–E4–G4, then E4–G4–C5, then G4–C5–E5, then C5–E5–G5. Then come back down. Say the name of each position as you play it.

4.2 Finding the root of any chord

When a chord is inverted, its root is no longer at the bottom. So how do you name a chord you find under your fingers? Look at the two gaps between its three notes.

Look at the numbers in Figure 4.1. In root position both gaps are thirds, 3 or 4 half steps. In each inversion one gap is wider, 5 half steps, and C sits directly above it.

Why 5? From the root up to the fifth is 7 half steps (C up to G). Once the root moves up an octave, the gap from the fifth up to it is the rest of that 12-key octave: G up to C is 5. So the root always lands right above the wider gap. That gives a rule:

> The note directly above the wider gap is the root.

Try it on A–D–F♯. From A up to D is 5 half steps: the wider gap. From D up to F♯ is 4. The note above the wider gap is D, so this is D major, in second inversion.

To name the position, look at the bottom note. With the third at the bottom (E–G–C) the chord is in first inversion; with the fifth at the bottom (A–D–F♯) it is in second inversion. If neither gap is wider than 4, the chord is in root position and its bottom note is the root.

4.3 Smooth chord changes: keep the shared note still

Now back to the question from the start of the chapter: how does the hand stay still? Two chords often share a note. C major (C, E, G) and F major (F, A, C) share C. C major and G major (G, B, D) share G. Keep the shared note under the same finger, and move the other fingers to the nearest notes of the new chord. Whatever inversion you end up in is the one you use.

Intuition

Think of the three notes of a chord as three singers in a choir. When the chord changes, no singer wants to leap. Each moves to the nearest note of the new chord, and anyone who already has a right note stays put.

Here is C to F, starting from C4–E4–G4 with fingers 1–3–5:

- the shared note is C, so your thumb stays on C4;
- E4 moves up one key to F4, and G4 moves up two keys to A4.

You now hold C4–F4–A4, which is F major in second inversion. Your hand stays where it was.

Now C to G, again from C4–E4–G4:

- the shared note is G, so your little finger stays on G4;
- E4 moves down two keys to D4, and C4 moves down one key to B3.

You now hold B3–D4–G4, which is G major in first inversion, played with fingers 1–2–5.

Put it together: C, F, G, C, the two versions from the start of the chapter. With root-position chords every finger moves at every change: added together, your fingers travel 42 keys. With the smooth version they travel 12. Figure 4.2 draws the two versions.

The same idea works for any progression. Here is I–V–vi–IV in C (C, G, Am, F) from Chapter 3, played smoothly:

- C: C4–E4–G4, fingers 1–3–5.
- G: B3–D4–G4, fingers 1–2–5. G stays.
- Am: C4–E4–A4, fingers 1–2–5. Every finger moves up a little.
- F: C4–F4–A4, fingers 1–3–5. C and A stay; only E moves up one key to F.

Hear it, ending back on C. Compare it with the jumpy version.

Common trap

Do not work out inversion names while you play. Find the shared note and the shortest move instead. The names are for talking about chords, not for finding them.

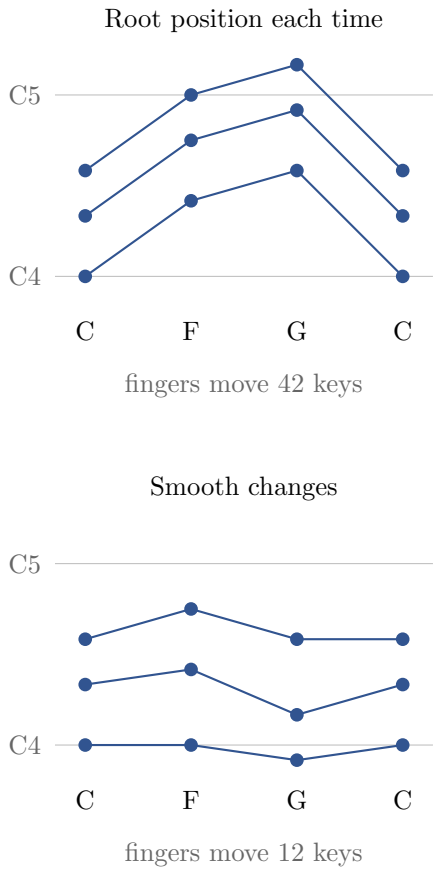


Figure 4.2: Each line follows one finger through C, F, G, C; higher means a higher key. At the top all three fingers jump; at the bottom they barely move.

4.4 Two hands: a bass note below, a chord above

Your right hand can now change chords smoothly near middle C. But in C4–F4–A4 the lowest note is C, not F. How does a listener hear an F chord? The left hand tells them by playing the root of the chord, low down. The lowest notes of a piece are called the **bass**.

The usual layout, shown in Figure 4.3:

- The left hand plays the root of each chord, low, between C2 and C3. It can play the root alone, the root with the note a fifth (7 half steps) above it, such as C2 and G2, or the root with the next root up, an octave, such as C2 and C3.
- The right hand plays the chord near middle C, using smooth inversions.

Here is I–V–vi–IV with the left hand on C3, G2, A2 and F2 under the smooth right-hand chords.

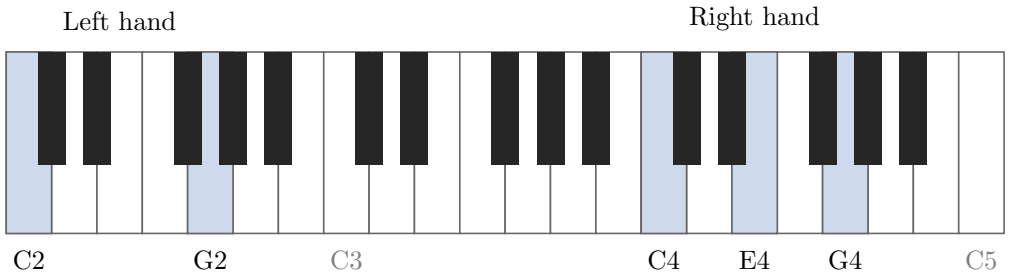


Figure 4.3: The left hand plays the root low (here C2 with G2). The right hand plays the chord near middle C.

Common trap

Do not play a full three-note chord low down in the left hand. Low notes close together sound thick and muddy. Compare a low C2–E2–G2 with a low C2–G2 under the same right hand. Below about C3, keep the left hand to single notes, fifths and octaves.

4.5 Three accompaniment patterns

Holding each chord for four counts works, but it sounds stiff. Playing chords to support a melody or a singer is called **accompaniment**, and a repeating pattern gives it life. Here are three patterns, each on the smooth I–V–vi–IV from above, with 4 counts on each chord. Count 1, 2, 3, 4 steadily and change chord each time you return to 1. (Chapter 5 covers rhythm in full.)

Block chords

Both hands play together on every count: left-hand root and right-hand chord, four times per chord. It is the plainest pattern, and the one to start with.

Broken chords

The right hand plays the chord one note at a time instead of all at once. On count 1, the left hand plays the root together with the right hand's bottom note, and holds it down for all 4 counts. Then the right hand goes on: middle note on 2, top note on 3, middle note on 4. It sounds like a rolling wave.

The pop bounce

The hands take turns. On 1, the left hand plays the low root and the right hand plays the chord, together. On 2, the right hand plays the chord alone. On 3, the left hand alone plays the fifth, 7 half steps above the root, with the thumb. On 4, the right hand plays the chord again.

Try it at the keyboard

Pick one pattern and play it on a single chord, C, until it feels comfortable. Then add the change to G, then Am, then F. Go slowly enough that you never have to stop. A steady slow pattern sounds better than a fast one with gaps.

4.6 Slash chords and stepwise bass lines

Song sheets sometimes show chords like C/E or G/B. The slash means: play the chord on the left, with the note on the right in the bass. So C/E means a C major chord with E as the lowest note. The right hand plays C major and the left hand plays E.

Why would a song ask for this? For the same reason the right hand uses inversions: so the bass can move smoothly too. Here is a common kind of opening: C, G/B, Am, Am/G, F. Look at the left-hand notes: C3, B2, A2, G2, F2. The bass walks down the white keys one at a time, while the right hand plays smooth chords on top. A bass that moves to a neighboring note each time, like this, is called a **stepwise bass line**; musicians also say the bass "walks down".

Work it yourself

Put your left thumb on C3. Fingers 2, 3, 4 and 5 then sit on B2, A2, G2 and F2. Put your right hand on C4–E4–G4. Play four slow counts on each chord:

- C: left C3, right C4–E4–G4;
- G/B: left B2, right B3–D4–G4;
- Am: left A2, right C4–E4–A4;
- Am/G: left G2, right C4–E4–A4 (the right hand does not move);
- F: left F2, right C4–F4–A4.

When it feels comfortable, try it with your eyes closed, and let the shared notes guide your hand.

What to remember

- An *inversion* is a chord with a note other than the root at the bottom. Notes an octave apart sound alike, so it is still the same chord.
- C major has three positions: root position C–E–G (fingers 1–3–5), first inversion E–G–C (1–2–5) and second inversion G–C–E (1–3–5).
- To find the root of an inverted chord, find the wider gap (5 half steps). The note directly above it is the root.
- For smooth chord changes, keep shared notes still and move the other fingers to the nearest notes.
- The *bass* is the lowest part. The left hand plays the root low (alone, as a fifth or as an octave); the right hand plays the chord near middle C. Avoid full chords low down.
- *Accompaniment* patterns: block chords, broken chords, and the pop bounce.
- A slash chord such as C/E means a C chord with E in the bass. Slash chords let the bass move smoothly in a *stepwise bass line*.

Exercises

Exercise 4.1. Write out the three positions of F major (F, A, C) and of G major (G, B, D), starting from F4 and G4. Give the right-hand fingers for each.

Answer

F major: root position F4–A4–C5 (fingers 1–3–5); first inversion A4–C5–F5 (1–2–5); second inversion C5–F5–A5 (1–3–5).
G major: root position G4–B4–D5 (1–3–5); first inversion B4–D5–G5 (1–2–5); second inversion D5–G5–B5 (1–3–5).
The same shapes also exist an octave lower, and those are the ones you will use most: for example C4–F4–A4 for F and B3–D4–G4 for G.

Exercise 4.2. Use the wider-gap rule to name each chord and its position: (a) F♯4–A4–D5, (b) E♭4–A♭4–C5, (c) B♭3–E♭4–G4.

Answer

(a) F♯ to A is 3 half steps; A up to D is 5, the wider gap. The note above it is D. The chord is D major (D, F♯, A) in first inversion, because the third, F♯, is at the bottom.
(b) E♭ up to A♭ is 5 half steps, the wider gap, so the root is A♭. The chord is A♭ major (A♭, C, E♭) in second inversion, with the fifth, E♭, at the bottom.
(c) B♭ up to E♭ is 5 half steps, so the root is E♭. The chord is E♭ major (E♭, G, B♭) in second inversion.

Exercise 4.3. In the key of G, I–V–vi–IV is G, D, Em, C. Start on G4–B4–D5 and find a smooth right-hand version of the whole progression. At each change, say which notes stay.

Answer

G to D (D, F♯, A): the shared note is D5, so it stays. G4 moves down one key to F♯4, and B4 moves down two keys to A4. You hold F♯4–A4–D5.
D to Em (E, G, B): no note is shared. Move each finger to the nearest note of Em: F♯4 up to G4, A4 up to B4, D5 up to E5. You hold G4–B4–E5.
Em to C (C, E, G): G4 and E5 stay. B4 moves up one key to C5. You hold G4–C5–E5.
Then back to G: G4 stays, C5 moves down to B4, and E5 moves down to D5.

Exercise 4.4. A song sheet shows G, D/F♯, Em. What do your two hands play for D/F♯? What do the three left-hand notes do?

Answer

For D/F♯, the right hand plays a D major chord (D, F♯, A), for example F♯4–A4–D5, and the left hand plays F♯. The left-hand notes for the three chords are G, F♯, E. For example G2, F♯2, E2: the bass walks down one step at a time, like the stepwise bass line in this chapter.

Part III

Rhythm and Reading Music

Chapter 5

Rhythm: Counting Time

Hum the notes of *Happy Birthday* with long, random pauses between them, and nobody will recognize the tune. Now tap its rhythm on a table with one finger, with no notes at all. Most people will name the song within a few taps.

That is how much of music lives in time. So far this book has been about which keys to press. This chapter is about when to press them, and for how long.

You do not need to read music for this chapter. The symbols for long and short notes appear here on their own, and you count them out loud. Chapter 6 puts them on the lines of written music.

5.1 The beat

Put on any song you like and tap your foot along with it. Your foot finds a steady pulse, like a clock ticking or a heart beating. That steady pulse is called the **beat**.

The beat keeps going at the same speed, even when the music pauses or a singer holds a long note. Here are eight steady beats played as clicks.

The speed of the beat is called the **tempo**. Musicians measure it in **beats per minute**, written BPM. At 60 BPM there is one beat every second, like a clock. At 120 BPM there are two beats every second, more like a brisk walk.

Listen to the same eight beats at 60 BPM and then at 120 BPM. The clicks are the same; only the speed changes.

A **metronome** is a small device or phone app that clicks a steady beat at any tempo you choose. It is the best rhythm teacher you can have, and many are free.

Try it at the keyboard

Set a metronome to 60 BPM, or play the first example above a few times. Tap your foot with the clicks. Then clap with them while you keep tapping. When that feels easy, try 90 BPM.

5.2 Rhythm: long and short sounds

The beat is steady. The notes on top of it are not: some are long and some are short. The pattern of long and short sounds is called the **rhythm**.

Sing the start of *Twinkle, Twinkle, Little Star* while you tap your foot. "Twin-kle twin-kle lit-tle" gets one tap for each part of a word. Then "star" lasts two taps. The beat never changed, but the last note was twice as long.

Here it is on the piano. Count "1, 2, 3, 4, 5, 6, 7–8" as you listen: the last note is held through two counts.

So beat and rhythm are two different things. The beat is the steady pulse you tap with your foot. The rhythm is the pattern your fingers play on top of it.

5.3 How long is each note?

The last note of *Twinkle* lasts two beats and the others one. How can a page say that? In written music, the word note also means the small symbol that stands for one sound, and its shape tells you how many beats to hold the key down.

A note symbol is built from up to three parts. The **notehead** is the oval. The **stem** is the thin line going up or down from it. A **flag** is the little curl on the end of some stems. Figure 5.1 shows the five note lengths you will meet most often.
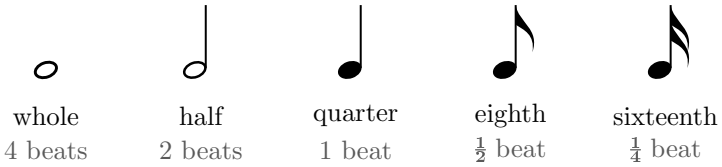


Figure 5.1: The five common note lengths. A hollow head means a long note; a filled head with more flags means a shorter one.

In most music, one quarter note lasts one beat, and this chapter assumes so until the section on time signatures. Each example below starts with four clicks to set the beat, called a **count-in**. The clicks then carry on under the notes, so you can hear how many beats each note lasts. Count along out loud.

- A **whole note** is a hollow oval with no stem. It lasts 4 beats.
- A **half note** is a hollow oval with a stem. It lasts 2 beats, so two of them take 4 beats.
- A **quarter note** is a filled oval with a stem. It lasts 1 beat: one note on every count.
- An **eighth note** has a filled oval, a stem and one flag. It lasts half a beat, so two fit in each beat.
- A **sixteenth note** has two flags. It lasts a quarter of a beat, so four fit in each beat.

Look at the names: they are fractions of the whole note. A half note is half of a whole note, a quarter note is a quarter of one, and an eighth note is an eighth. Each step down the list cuts every note

into two equal pieces.

Figure 5.2 shows the cutting. Read it from the top down: one whole note splits into two halves, each half into two quarters, each quarter into two eighths. Every row still fills the same four beats.

Why halves? Because halving keeps every note lined up with the beat. Two eighth notes fill one beat exactly, and so do four sixteenths, so you can always count them against your foot.

In the bottom row of the figure the eighth notes come in pairs, with their flags joined into one thick line. That line is called a **beam**. Two beamed eighth notes look like a little bridge and together last one beat. Sixteenth notes are joined by two beams.

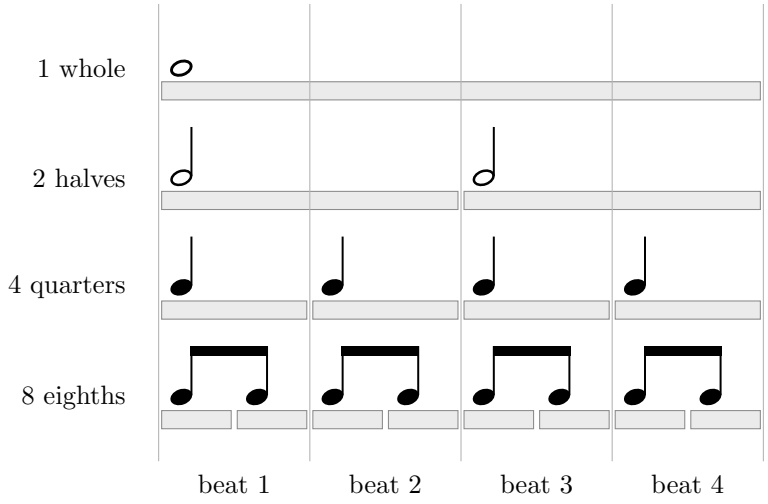


Figure 5.2: Each row fills the same four beats. Going down one row splits every note into two notes half as long.

5.4 Rests: counting the silence

Music needs silence as well as sound. A **rest** is a symbol that means "play nothing here, and keep counting". Each note length has a matching rest, shown in Figure 5.3.



Figure 5.3: The four common rests. The whole rest hangs down from a line; the half rest sits on top of a line, like a hat.

- A **whole rest** is a small black block hanging below a line. It means a whole bar of silence (we meet bars in the next section). Here is a whole note, a whole rest, and another whole note.
- A **half rest** is the same block sitting on top of a line. It lasts 2 beats. A half note, a half rest, then the same again.

- A **quarter rest** looks like a squiggle standing up. It lasts 1 beat. Note, rest, note, rest.
- An **eighth rest** looks like a small 7 with a dot. It lasts half a beat.

To tell the whole and half rests apart, remember: the whole rest is heavier, so it hangs down; the half rest is lighter, so it sits up.

Common trap

Beginners often rush through rests, as if silence did not count. It does. Keep tapping your foot and counting out loud through every rest, so the next note comes in exactly on time.

5.5 Bars and time signatures

Listen to these clicks. Every fourth click is higher, and your ear starts grouping them in fours without being asked. Most music works like this: a stronger beat comes back at a steady distance, with weaker beats in between.

Written music marks these groups. Each group of beats is a **bar** (also called a **measure**), and a thin vertical line called a **bar line** marks where one bar ends and the next begins. A double bar line, thin then thick, marks the end of the piece.

How does a player know whether to group in fours or in threes? The music says so at the very start, with two numbers stacked on top of each other. This is the **time signature**. In this book we write it on one line, like "4/4" (say "four four").

- The top number says how many beats are in each bar.
- The bottom number says which note gets one beat. A 4 means the quarter note gets one beat; an 8 means the eighth note does.

Figure 5.4 shows four bars in 4/4 time. Look at how each bar is filled differently, but every bar adds up to exactly four beats.

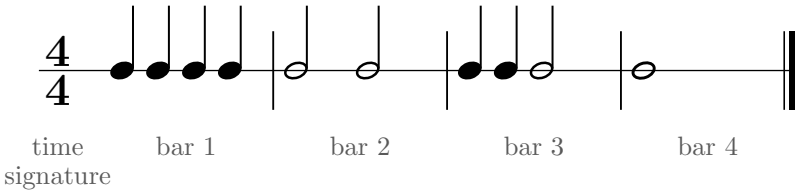


Figure 5.4: Four bars of 4/4 time. Every bar adds up to four beats: four quarters, two halves, two quarters and a half, or one whole note.

4/4: four beats in a bar

Count "**1**, 2, 3, 4, **1**, 2, 3, 4" and lean a little on each 1, as in the clicks at the start of this section.

Most pop, rock and folk songs are in 4/4. It is so common that it is also called **common time**, and is sometimes written as a large letter C instead of two numbers.

3/4: three beats in a bar

Count "**1**, 2, 3, **1**, 2, 3". This is the swinging feel of a waltz.

Happy Birthday and *Amazing Grace* are both in 3/4. Hum one of them and count along: the strong "1" keeps coming back every three beats.

6/8: two groups of three

In 6/8 there are six eighth notes in each bar, grouped as three and three. Count "**1** 2 3 **4** 5 6". Your foot taps only twice per bar, on 1 and on 4, so it feels like a gentle rocking. *Row, Row, Row Your Boat* is usually written in 6/8.

Both 3/4 and 6/8 can hold six quick notes in a bar, so what is the difference? The grouping. In 3/4 the six notes come in three pairs, "**1** and **2** and **3** and", and your foot taps three times. In 6/8 they come in two threes, "**1** 2 3 **4** 5 6", and your foot taps twice. Same number of notes, a different swing. Figure 5.5 shows both: look at how the beams group the notes.

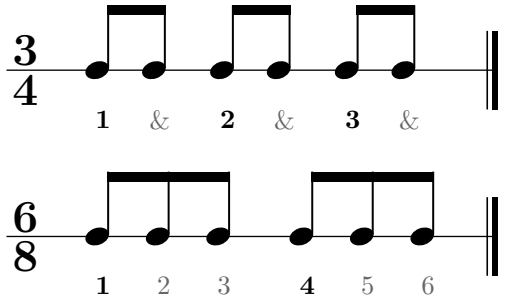


Figure 5.5: Six eighth notes in a bar of 3/4 and in a bar of 6/8. In 3/4 they are beamed in three pairs; in 6/8, in two groups of three. The bold counts are where your foot taps.

5.6 Counting out loud

Your foot can mark the beats, but an eighth note falls halfway between two taps. How do you place it? Count out loud while you play. Say the beat numbers on the beats, and between them say "and", exactly halfway to the next beat:

1 and 2 and 3 and 4 and

On paper this is written "1 & 2 & 3 & 4 &".

Quarter notes fall on the numbers. Eighth notes fall on the numbers and on every "and". Listen to four quarter notes and then eight eighth notes, and say "1 and 2 and 3 and 4 and" with both.

Now mix them. Try a quarter, two eighths, a quarter and a quarter. You play on "1, 2 and, 3, 4".

For sixteenth notes, split each beat into four and say "1 e and a" ("one ee and uh"). You will not need these much at first.

Try it at the keyboard

Clap each pattern while counting "1 and 2 and 3 and 4 and" out loud. Start slowly.

1. Clap on every number: four quarter notes.
2. Clap on every number and every "and": eight eighth notes.
3. Clap on 1 and 3 only, holding your hands together until the next clap: two half notes.

5.7 Dotted notes

A bar of 3/4 holds three beats. How would you write one note that fills it? A half note is too short (2 beats) and a whole note too long (4). Written music solves this with a dot.

A small dot right after a note adds half of that note's own length.

- A **dotted half note** lasts 2 + 1 = 3 beats, exactly one bar of 3/4.
- A **dotted quarter note** lasts 1 + a half = one and a half beats.

The dotted quarter is the one that confuses people, and counting out loud fixes it. It is almost always followed by an eighth note, and together they fill two beats. Hold the dotted quarter through "1 and 2", then play the eighth note on the "and" after 2. Figure 5.6 shows this: the brackets under the counts show how long each note lasts.

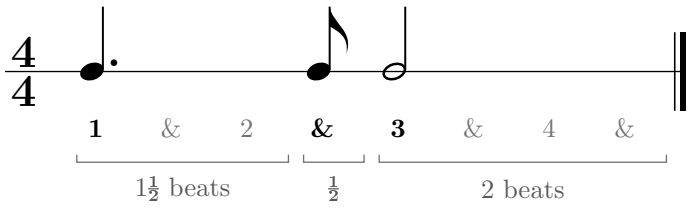


Figure 5.6: A dotted quarter, an eighth and a half note. The bold counts are the ones you play on: 1, the "and" after 2, and 3.

Listen to that bar: you hear notes on 1, on the "and" of 2, and on 3.

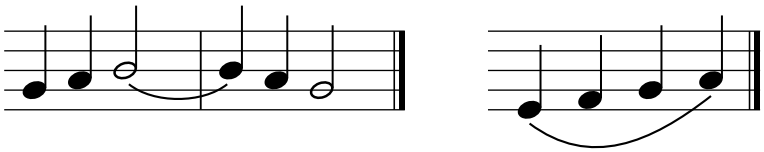
You know this rhythm already. In the chorus of *Jingle Bells*, "jin-gle all the way" uses it: "all" is a dotted quarter and "the" is an eighth.

5.8 Ties and slurs

A dot can make a note longer, but only within its own bar. What if a note has to last past the bar line? A **tie** is a curved line joining two notes of the same pitch. Play the first note and hold it down for both lengths added together. Do not play the second one again.

In Figure 5.7 (left), a half note at the end of the first bar is tied to a quarter note at the start of the next, so you play once and hold for 2 + 1 = 3 beats. Look at the two tied notes: they sit at the same height, because they are the same key. Listen: G, A, then B held for three beats, then A and G.

A curved line joining notes of different pitches looks similar but means something else (Figure 5.7, right). It is a **slur**, and it asks you to play those notes smoothly, with no gap between them. Musicians call this smooth, connected playing **legato**.



tie: play once,
hold 2 + 1 = 3 beats

slur: different notes,
play them smoothly

Figure 5.7: Left: a tie joins two notes of the same pitch across a bar line. Right: a slur arches over notes of different pitches.

Pause

Same pitch under the curve: it is a tie, so hold the note. Different pitches under the curve: it is a slur, so play each note, smoothly connected.

5.9 Syncopation

Most of the time, important notes land on the beat. Sometimes a note arrives on an "and" instead and the beat right after it is skipped. That surprise is called **syncopation**, and it gives music a bounce.

Count "1 and 2 and 3 and 4 and" and play on 1, on the first "and", on the next "and", then on 3 and 4. Beat 2 is never struck; the note before it is held through it.

You hear syncopation all the time in pop, jazz, funk and ragtime. When a rhythm feels tricky, write the counts under it and clap it slowly.

5.10 Two hands, one rhythm

Playing different rhythms with two hands sounds hard. The trick is to stop thinking of two separate rhythms and make them into one.

Write the counts "1 & 2 & 3 & 4 &" in a row. Under each count, mark what happens:

- T for together: both hands play.
- R for right hand only.
- L for left hand only.
- a dash (–) when nothing new is played.

Here is an example. The left hand plays two half notes, on 1 and 3. The right hand plays a quarter, two eighths and a half note, on 1, 2, the "and" of 2, and 3.

count	1	&	2	&	3	&	4	&
right hand	x		x	x	x			
left hand	x				x			
together	T	–	R	R	T	–	–	–

Now the two hands are one pattern: "T, –, R, R, T, –, –, –". Tap it on your knees while counting out loud, right hand on your right knee and left hand on your left. Then try it at the piano. In this recording the left hand plays C3 and the right hand plays E, D, E, C.

Try it at the keyboard

Tap the pattern above on your knees ten times, slowly, counting out loud. Then speed up a little. If it falls apart, slow down again: slow and steady always beats fast and shaky.

What to remember

- The *beat* is the steady pulse you tap your foot to. The *tempo* is its speed, in beats per minute (BPM).
- The *rhythm* is the pattern of long and short notes played over the beat.
- Note lengths: whole = 4 beats, half = 2, quarter = 1, eighth = half a beat, sixteenth = a quarter of a beat.
- A *rest* is a measured silence. Each note length has a matching rest. Keep counting through rests.
- *Bars* split the music into equal groups of beats. The *time signature* (4/4, 3/4, 6/8) tells you how many beats are in each bar.
- Count out loud: "1 and 2 and 3 and 4 and". A dot adds half a note's length; a *tie* joins two notes into one long one.
- *Syncopation* puts notes on the "and" and skips the beat. For two hands, combine them into one T/R/L pattern.

Exercises

Exercise 5.1. How many quarter notes last as long as one whole note? How many eighth notes last as long as one half note?

Answer

A whole note lasts 4 beats and a quarter note lasts 1, so four quarter notes fill one whole note. A half note lasts 2 beats and each eighth note lasts half a beat, so four eighth notes fill one half note. You can check both in Figure 5.2.

Exercise 5.2. A bar of 4/4 starts with a half note and a quarter note. What single note would complete the bar? What single rest would complete it?

Answer

The half note (2 beats) and the quarter note (1 beat) use 3 of the 4 beats. One beat is left, so a quarter note completes the bar. A quarter rest would also complete it, with one beat of silence at the end.

Exercise 5.3. In 4/4, a bar has a dotted quarter note, an eighth note and a half note. Write "1 & 2 & 3 & 4 &" and circle the counts where you play.

Answer

You play on 1, on the "&" after 2, and on 3. The dotted quarter is held through "1 & 2" (one and a half beats). The eighth note takes the "&" of 2. The half note starts on 3 and is held through "3 & 4 &".

Exercise 5.4. Six quick notes can be grouped as 3/4 or as 6/8. In your own words, how does each one feel when you tap your foot?

Answer

In 3/4 the six notes come in three pairs: "1 and 2 and 3 and", and your foot taps three times per bar. In 6/8 they come in two groups of three: "1 2 3 4 5 6", and your foot taps only twice, on 1 and 4. The 3/4 version feels like a waltz; the 6/8 version feels like rocking in a boat.

Exercise 5.5. In one bar of 4/4, the left hand plays half notes on 1 and 3. The right hand plays a quarter note, two eighth notes, a dotted quarter note and an eighth note. Fill in the T/R/L/– row for "1 & 2 & 3 & 4 &".

Answer

First find where the right hand plays. The quarter note is on 1. The two eighths are on 2 and the "&" of 2. The dotted quarter starts on 3 and lasts one and a half beats, through "3 & 4". The last eighth note falls on the "&" of 4. The left hand plays on 1 and 3.
So the combined row is: T, –, R, R, T, –, –, R. Tap it on your knees, then listen.

Chapter 6

Reading Notes on the Staff

A note symbol from Chapter 5 says how long to play, but not which key. How could a page show which key? Put higher notes higher on the page. Written music does exactly that: the page becomes a picture of pitch, where up means higher in sound and further right on the keyboard.

You will not need to memorize every note. You will learn a few landmark notes and read the rest by counting small steps from them, the way you find a house from the nearest street corner.

6.1 The staff: a ladder of lines and spaces

Music is written on five horizontal lines called a **staff**. The five lines leave four spaces between them. We count both from the bottom: line 1 is the bottom line, and space 1 is the space directly above it.

A notehead can sit *on a line*, with the line running through its middle. Or it can sit *in a space*, between two lines. Figure 6.1 shows nine notes climbing the staff: line, space, line, space, and so on.

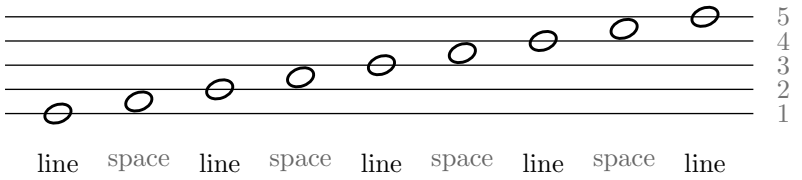


Figure 6.1: A staff has five lines and four spaces. Each note here is one step higher than the one before: line, space, line, space.

Each line and each space stands for one white key. Moving up from a line to the very next space (or from a space to the next line) moves up one white key, to the next letter of the musical alphabet. We call that move one **step**.

So the nine notes in Figure 6.1 are nine white keys in a row, and climbing the staff is the same as walking up the white keys: C, D, E, F, G, A, B, C, and so on. Using the spaces as well as the lines packs nine white keys into five lines that the eye can still tell apart.

A step on the staff counts letters, not keys: it can be a half step on the keyboard (E to F, B to C) or a whole step (C to D). The black keys do not get their own lines or spaces. They are written with

sharp and flat signs, shown later in this chapter.

6.2 Clefs: which note is which

The staff is a ladder of white keys, but which rung is which? Five lines could sit anywhere among the 52 white keys of a piano. A **clef**, a symbol drawn at the start of every staff, pins one line to one known note. Every other line and space follows by counting steps from it.

A piano covers more than seven octaves, far more than one staff of nine notes. So piano music uses two clefs: one for a staff of high notes, one for a staff of low notes.

The **treble clef** is the tall, curly sign. Its inner curl wraps around line 2, and marks that line as G4, the first G above middle C. The sign grew out of a fancy letter G, which is why it is also called the G clef. The treble staff holds higher notes, usually played by the right hand.

The **bass clef** looks like a backwards C with two dots. The two dots sit on either side of line 4, and mark that line as F3, the first F below middle C. It grew out of a letter F, so it is also called the F clef. The bass staff holds lower notes, usually played by the left hand.

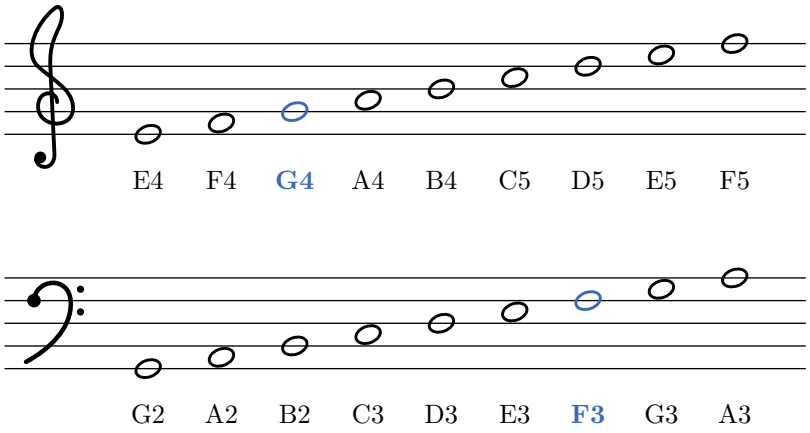


Figure 6.2: The treble clef marks G4 (line 2) and the bass clef marks F3 (line 4). Every other line and space follows by stepping up or down.

From those two notes you can fill in the rest by stepping, as in Figure 6.2.

- Treble staff, lines from the bottom: E4, G4, B4, D5, F5. Spaces: F4, A4, C5, E5.
- Bass staff, lines from the bottom: G2, B2, D3, F3, A3. Spaces: A2, C3, E3, G3.

You may have heard rhymes for these, like “Every Good Boy Does Fine” for the treble lines and “FACE” for the treble spaces. They work, but reciting a rhyme for every note is slow. The landmark method below is quicker once it clicks, and you can always fall back on a rhyme.

6.3 The grand staff and middle C

Piano music uses two staves at once, joined at the left by a long line or a curly bracket. Together they are called the **grand staff**: treble on top for the right hand, bass below for the left hand.

The two staves fit together like one tall ladder. The top line of the bass staff is A3. One step up is B3, in the space directly above it. One more step up is middle C, C4. One more step is D4, in the space directly below the treble staff. And one more step lands on E4, the bottom line of the treble staff.

Middle C sits between the two staves, so it gets its own short line drawn through it. A short line like this, added above or below a staff, is called a **ledger line**. Figure 6.3 shows middle C on its ledger line, halfway between the staves.

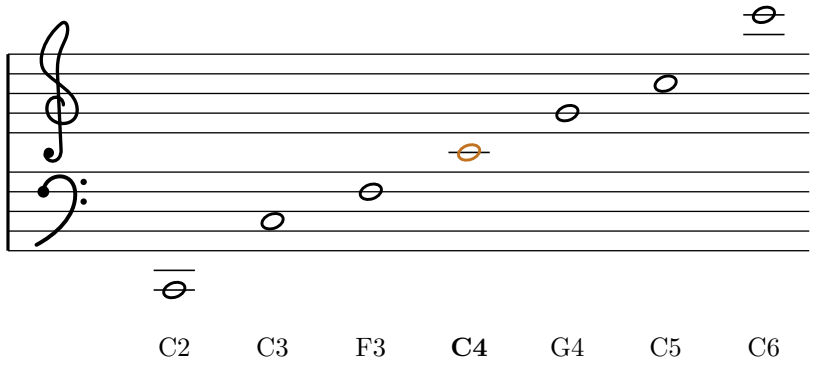


Figure 6.3: The grand staff, drawn as one ladder. Middle C (C4) sits on its own ledger line between the staves. The other notes shown are the landmarks of the next section.

In printed music the two staves are drawn further apart, to leave room. Then middle C appears in one of two places: on a ledger line directly below the treble staff when the right hand plays it, or directly above the bass staff when the left hand plays it. Both are the same key, as Figure 6.4 shows.

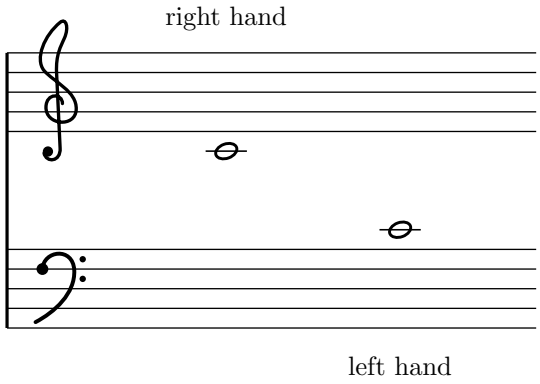


Figure 6.4: In printed music, middle C is written directly below the treble staff (for the right hand) or directly above the bass staff (for the left hand). Both mean the same key.

6.4 Ledger lines

What about notes higher or lower than a staff can hold? Use the same trick as middle C: add short lines. Each ledger line is one more line of the staff, and the spaces between them count too.

Above the treble staff: the top line is F5. The space above it is G5, then A5 sits on the first ledger line, B5 above it, and C6 on the second ledger line. Below the treble staff: D4 hangs directly under

the bottom line, then C4 on one ledger line, then B3 and A3.

The bass staff works the same way. Below its bottom line (G2) come F2, then E2 on the first ledger line, D2, and C2 on the second ledger line.

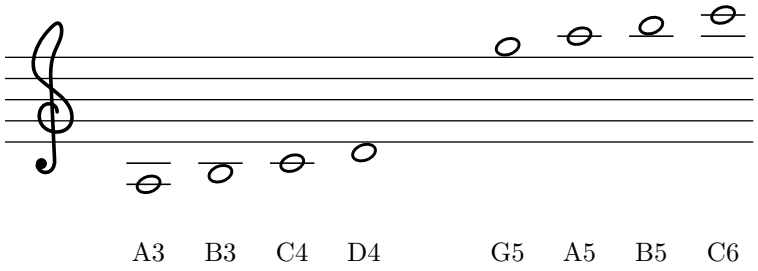


Figure 6.5: Ledger lines extend the treble staff downward (A3 to D4) and upward (G5 to C6). Count them like ordinary lines and spaces.

6.5 Landmark notes

Reading music fluently does not mean naming every note from scratch. It means knowing a handful of landmark notes by sight, and counting a step or two from the nearest one.

Here are five landmarks, all around middle C (they are in Figure 6.3):

- Middle C (C4), on its own ledger line between the staves.
- G4, the treble clef line (line 2 of the treble staff).
- F3, the bass clef line (line 4 of the bass staff).
- C5, in space 3 of the treble staff.
- C3, in space 2 of the bass staff.

Look for the mirror pattern in Figure 6.3. G4 is the second line up from the bottom of the treble staff; F3 is the second line down from the top of the bass staff. Both sit four steps away from middle C, one above and one below. C5 and C3 sit seven steps away, again one above and one below.

Hear the five landmarks from low to high.

From C3 up to C5, every white key is at most two steps from one of these five landmarks. To read a note, find the nearest landmark and count. One step goes from a line to the next space (or a space to the next line). Two steps go from a line to the next line, or from a space to the next space.

For notes further out, add four more landmarks: the bottom line of the bass staff (G2), the top line of the treble staff (F5), and the two Cs on second ledger lines, C2 below the bass staff and C6 above the treble staff. With all nine, every white key from C2 up to C6 is within two steps of a landmark.

A few examples:

- The next treble line above G4 is two steps up (G, A, B), so it is B4.
- The space directly above middle C is one step up, so it is D4.
- The next bass line below F3 is two steps down (F, E, D), so it is D3.

Try it at the keyboard

Put the thumb of your right hand on middle C and look at Figure 6.3. Play C4, then G4 with your little finger. Then put your left little finger on C3 and play F3 with your index finger (finger 2), then C3 again. Say each landmark's name out loud as you play it.

6.6 Sharps, flats and naturals on paper

A black key has no line or space of its own. So it borrows the position of a white key next to it, with a sign written directly before the note, on the same line or space.

- A **sharp** sign (♯) means: play the key one half step higher. A note on the F4 space with a sharp is F♯4, the black key directly right of F4.
- A **flat** sign (♭) means: play the key one half step lower. A note on the B4 line with a flat is B♭4, the black key directly left of B4.
- A **natural** sign (♮) cancels a sharp or flat and brings the note back to the white key.

A sharp, flat or natural written in front of a note is called an **accidental**. It lasts until the end of the bar, for every note on that same line or space. At the next bar line it is forgotten.

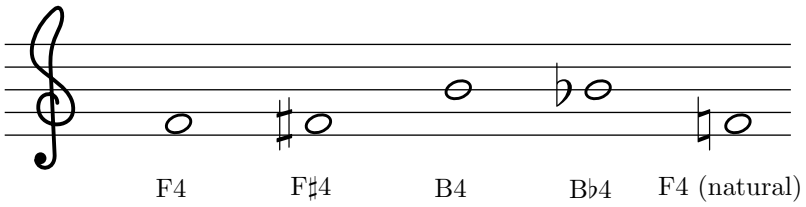


Figure 6.6: A sharp raises a note by a half step, a flat lowers it by a half step, and a natural cancels either one.

6.7 Key signatures

Chapter 2 showed that the scale of G major uses F♯ instead of F. A piece in the key of G (a piece built on that scale) would need a sharp in front of every F. Writing them all out would be tiring.

Instead, the sharps or flats a piece needs are written once, at the start of every staff, directly after the clef. This group is called the **key signature**. A sharp on the F line in the key signature means: every F in the piece, in every octave, is played as F♯, unless a natural sign says otherwise.

Figure 6.7 shows four key signatures. Look at where each sign sits: the sharp for F♯ is on the top line, which is an F, and the flat for B♭ is on the middle line, which is a B.

The signs always come in the same order. Each step clockwise round the circle of fifths in Chapter 2 adds one sharp, and the sharps arrive in the order F, C, G, D, A, E, B. Each step the other way adds one flat, in the reverse order: B, E, A, D, G, C, F.

Two quick tricks tell you the key:

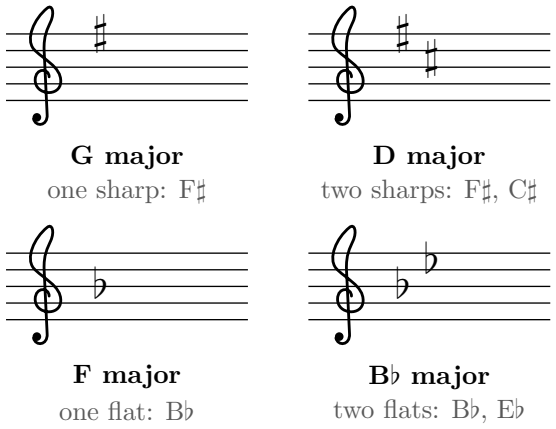


Figure 6.7: Four key signatures on the treble staff. The signs sit on the lines and spaces of the notes they change.

- *Sharps:* take the last sharp and go up one half step. One sharp (F♯): up a half step is G, so G major. Two sharps (F♯, C♯): the last is C♯, up a half step is D, so D major.
- *Flats:* the second-to-last flat is the key. Two flats (B♭, E♭): the second-to-last is B♭, so B♭ major. One flat has no second-to-last, so learn that one by heart: one flat (B♭) means F major.

No sharps or flats at all means C major.

Each key signature is also shared by one minor key, its relative minor from Chapter 2. To find it, count down three half steps from the major key's name. G major and E minor share one sharp; C major and A minor share no sharps or flats. The last note of a piece usually tells you which one it is: a piece in E minor tends to end on E.

6.8 Intervals by shape

In Chapter 1 you measured an *interval*, the distance between two notes, in half steps. On the staff it is named by counting letters, including both ends. C up to E is a third, because it covers three letters: C, D, E.

The staff shows intervals as shapes, so you can read them without naming the notes. Figure 6.8 shows the common ones, all starting from E4 on the bottom line.

- A step (a second) goes from a line to the very next space, or a space to the next line.
- A **skip** (a third) goes from a line to the next line, or a space to the next space.
- A fourth goes from a line to the second space above it.
- A fifth goes from a line to a line two away, skipping one line in between.
- An octave (an eighth) goes from E to the next E, from a line to a space.

A pattern helps here. Odd intervals (third, fifth, seventh) go from a line to a line, or a space to a space. Even intervals (second, fourth, sixth, octave) switch between line and space.

This links directly to your fingers. With one finger on each of five neighboring white keys, a step

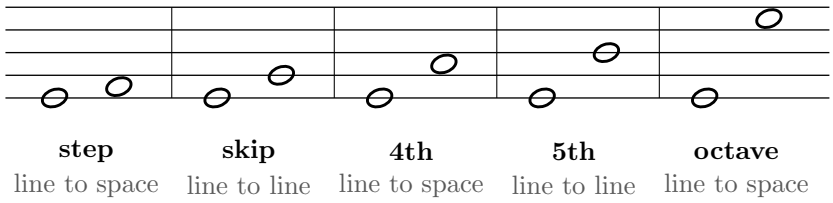


Figure 6.8: Intervals as shapes: a step, a skip, a fourth, a fifth and an octave, each starting on the bottom line (E4 on the treble staff).

moves to the next finger, a skip skips one finger, and a fifth goes from thumb to little finger. Once your hand is in place, you can read a melody by its ups and downs alone.

6.9 Chord shapes on the staff

A chord written on the staff is a stack of noteheads, one above the other, to be played together. The three positions of a chord from Chapter 4 each have their own shape, as Figure 6.9 shows.

- *Root position* (C–E–G) stacks on three lines in a row, or three spaces in a row. It looks like a snowman.
- *First inversion* (E–G–C) has two notes close together at the bottom and a wider gap on top.
- *Second inversion* (G–C–E) has a wider gap at the bottom and two notes close together on top.

The gap tells you where the root is. In both inversions, the root (the note the chord is named after) is the note directly above the gap.

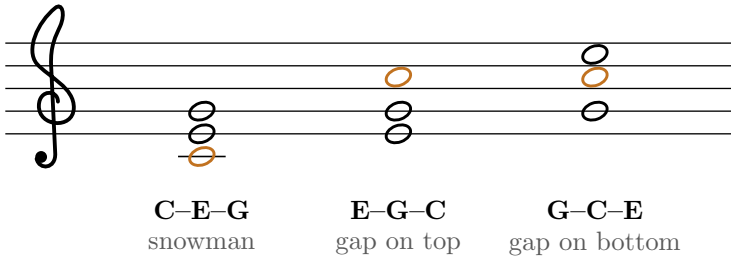


Figure 6.9: The C major chord in its three positions. The root C is colored; in each inversion it sits directly above the gap.

6.10 Calm habits for reading new music

Reading a piece you have never seen is called **sight-reading**. Here are five habits that make it calmer.

1. *Look before you play.* Spend half a minute on the page first. Check the key signature (which black keys are in use) and the time signature (how to count). Find the first note for each hand and put your hands in place.

2. *Pick a slow tempo and keep the beat.* Choose a speed at which the hardest bar feels easy. Count one bar out loud before you start.
3. *Look a little ahead.* While you play one note, your eyes should already be on the next. Try not to look down at your hands; feel for the black keys instead.
4. *Do not stop for wrong notes.* A missed note is forgotten in a moment; stopping breaks the music. Keep the beat and land on the next note.
5. *Practice slowly.* It is better to play four bars slowly and steadily than fast with stops. Speed comes by itself once the notes feel familiar.

What to remember

- A *staff* has five lines and four spaces. Each line or space is one white key; moving up one *step* (line to next space) moves up one letter.
- The *treble clef* marks G4 on line 2; the *bass clef* marks F3 on line 4. Together they make the *grand staff*.
- Middle C sits on a short *ledger line* between the staves. Ledger lines extend any staff up or down.
- Landmarks: C3, F3, C4, G4 and C5 (plus G2, F5, C2 and C6 further out). Read other notes by counting steps from the nearest one.
- A *sharp* raises a note a half step, a *flat* lowers it, a *natural* cancels either.
- The *key signature* lists the sharps or flats used all through a piece. Sharps: last sharp up a half step. Flats: second-to-last flat.
- *Intervals* have shapes: step = line to space, skip = line to line. Chords show as a snowman, gap on top, or gap on bottom; the root sits directly above the gap.

Exercises

Exercise 6.1. Use the landmarks. Name the note (a) on the next treble line above G4, (b) in the space directly below middle C, (c) on the next bass line below F3.

Answer

(a) From G4, the next line up is two steps higher: G, A, B. It is B4.
(b) From middle C, one step down is B3, which sits in the space directly under middle C's ledger line.
(c) From F3, the next line down is two steps lower: F, E, D. It is D3.

Exercise 6.2. Name the note on the first ledger line above the treble staff, and the note on the second ledger line below the bass staff.

Answer

The top line of the treble staff is F5. One step up is G5 (in the space above), and one more step is the first ledger line: A5.
The bottom line of the bass staff is G2. Stepping down: F2 (space), E2 (first ledger line), D2 (space), C2 (second ledger line). It is C2, one of the far landmarks.

Exercise 6.3. A key signature has three sharps. Which major key is it? Another has three flats. Which major key is that?

Answer

Sharps always come in the order F, C, G, so three sharps are F♯, C♯ and G♯. The last one is G♯; one half step up is A. The key is A major.
Flats come in the order B, E, A, so three flats are B♭, E♭ and A♭. The second-to-last is E♭, so the key is E♭ major.

Exercise 6.4. On the treble staff, one note sits on the bottom line and another on the middle line. What is the interval between them? Which fingers would play it in a five-finger position starting on the lower note?

Answer

Line to line, with one line skipped in between, is a fifth. The notes are E4 and B4: E, F, G, A, B is five letters. With your thumb on E4, the fifth is under your little finger: fingers 1 and 5.

Exercise 6.5. Three notes are stacked on the treble staff: E4 (line 1), A4 (space 2) and C5 (space 3). Which shape is this: snowman, gap on top, or gap on bottom? What is the chord?

Answer

From E4 to A4 is a fourth, a wider gap; from A4 to C5 is a skip. So the gap is on the bottom, and this is a second inversion. The root is the note directly above the gap: A. The notes A, C and E make an A minor chord, here played as E–A–C, with fingers 1, 3 and 5.

Part IV

Playing Real Music

Chapter 7

Your First Pieces

You now know where the notes are, how scales and chords are built, how to count rhythm and how to read the staff. This chapter puts them together in real music.

There are three pieces, learned in four steps. Each step adds one new thing to the last:

- *Ode to Joy*, right hand: five fingers on five keys, one note per beat.
- *Ode to Joy* with the left hand: first single bass notes, then the chords from Chapters 3 and 4.
- *Amazing Grace*: 3/4 time, a note before the first bar, a key signature and a small hand move.
- Four chords that fit hundreds of pop songs: an accompaniment with no melody at all.

Both melodies are old enough to be free for anyone to play.

Each piece is written three ways: as letters with finger numbers, on the staff, and as sound. Use whichever helps most, and let the others check your work.

7.1 Getting ready: the C position

Most first pieces use one hand position, so your fingers never have to search. In the C position from Chapter 1, each finger of the right hand rests on one white key, starting from middle C: thumb (finger 1) on C4, then D4, E4, F4, and little finger (finger 5) on G4.

The left hand mirrors this, an octave lower. Little finger (5) on C3, then 4 on D3, 3 on E3, 2 on F3, and thumb (1) on G3. Figure 7.1 shows both hands.

Sit so your forearms are about level with the keys. Curve your fingers gently, as if holding a ball, and play with the fleshy tips. Keep your shoulders loose.

Try it at the keyboard

Place your right hand in C position. Play C, D, E, F, G and back down, saying the finger numbers out loud: 1, 2, 3, 4, 5, 4, 3, 2, 1. Do the same with the left hand: 5, 4, 3, 2, 1, 2, 3, 4, 5.

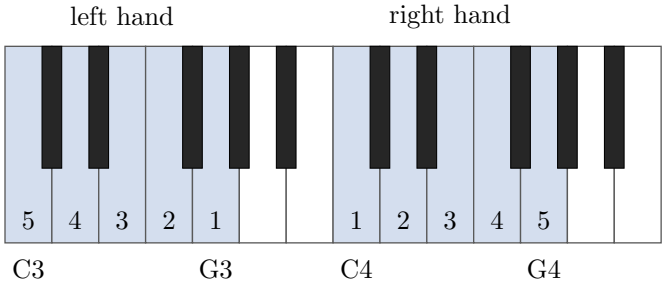


Figure 7.1: The C position. The numbers are the fingers that rest on each key: thumbs are 1, little fingers are 5.

7.2 Ode to Joy: the right hand

Beethoven wrote this tune for the last movement of his Ninth Symphony, in 1824. It moves mostly by steps, so the whole tune fits under a hand in C position: the five keys you have been playing are all it needs. Its time signature is 4/4, and almost every note is a quarter note: one note per beat.

The tune has two phrases. A **phrase** is a musical sentence, a few bars long, that ends with a feeling of a pause. Here is the first one, bar by bar:

	bar 1	bar 2	bar 3	bar 4
notes	E E F G	G F E D	C C D E	E. D D
fingers	3 3 4 5	5 4 3 2	1 1 2 3	3 2 2

Bars 1 to 3 are four quarter notes each. Bar 4 is the rhythm from Chapter 5: a dotted quarter (the "E."), an eighth note, and a half note. Count "1 and 2 and 3 and 4 and": E on 1, the first D on the "and" after 2, and the last D on 3, held to the end of the bar.

Listen to the first phrase. It ends on D, which sounds a little unfinished, like a question.

The second phrase starts the same way. Only the last bar changes, and it ends on C, which sounds like an answer:

	bar 5	bar 6	bar 7	bar 8
notes	E E F G	G F E D	C C D E	D. C C
fingers	3 3 4 5	5 4 3 2	1 1 2 3	2 1 1

Listen to the second phrase, then to both phrases together.

Figure 7.2 shows the same melody on the treble staff, with finger numbers above the notes and letters below. Look at how the noteheads climb and fall one step at a time, line to space to line, and how each step up the staff is one finger further along.

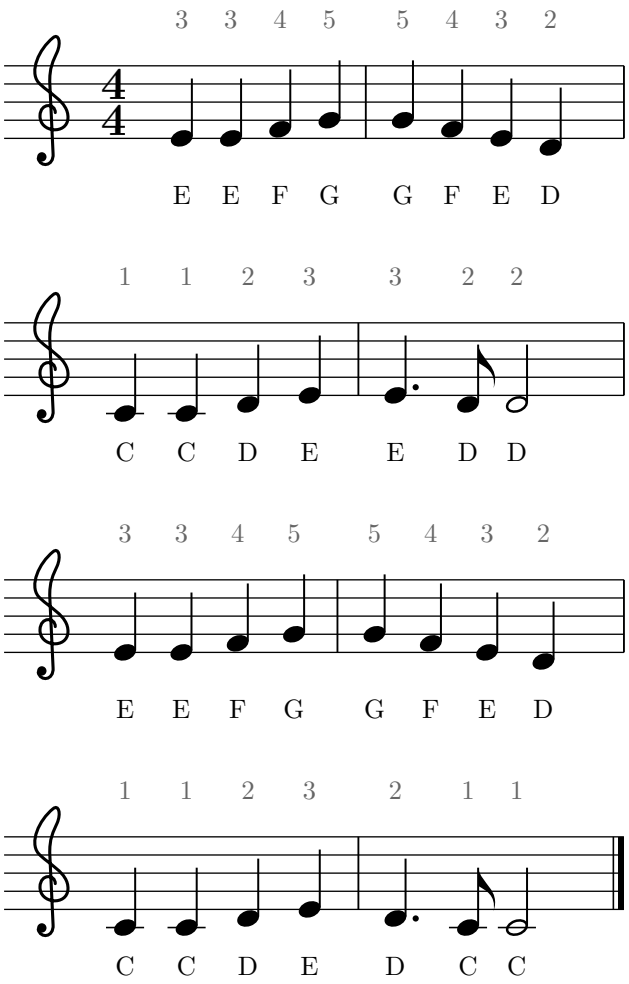


Figure 7.2: Ode to Joy, both phrases, in C position. Finger numbers are above the staff and note names below.

Try it at the keyboard

Play the first phrase slowly, at about half the speed of the recording, counting out loud. If you play a wrong note, keep going to the end. Then play it three more times. When it feels steady, learn the second phrase, which only changes in the last bar.

7.3 Ode to Joy: adding the left hand

Now give the melody a floor to stand on. We add the left hand in two stages: first single bass notes, then full chords.

Stage 1: bass notes

Put your left hand in C position. You need only two keys: C3 under your little finger (5) and G3 under your thumb (1). C goes with the C major chord and G with the G major chord, the I and V chords of the key of C from Chapter 3.

How do you know which one fits a bar? Look at the melody. Bar 1 (E, E, F, G) leans on E and G, two notes of the C chord, so C goes under it. Bar 2 (G, F, E, D) leans on G and D, two notes of the G chord (G, B, D), so G goes under it.

Play one bass note at the start of each bar, together with the first melody note, and hold it while the right hand carries on. In the last bar of each phrase, the bass changes halfway, on beat 3:

bar	1	2	3	4	5	6	7	8
left hand	C	G	C	C, then G	C	G	C	G, then C

Listen to the first phrase with bass notes and the second.

Practice each hand alone first. Then put them together slowly, using the T/R/L idea from Chapter 5: on the first beat of each bar both hands play (T), and on the other beats only the right hand plays (R). The one exception is beat 3 of each phrase's last bar, where the bass changes: that beat is T as well.

Stage 2: chords

When the bass notes feel easy, replace them with three-note chords. The C chord is C3–E3–G3, with fingers 5, 3 and 1.

For the G chord, use the smooth change from Chapter 4. Keep your thumb on G3. Move your middle finger down one white key, from E3 to D3, and your little finger down a half step, from C3 to B2. That gives B2–D3–G3, a G chord in first inversion, with fingers 5, 3, 1. Your hand hardly moves: C chord, G chord, C chord.

Play each chord where the table above has its bass note. Here is the first phrase with chords and the second.

7.4 Amazing Grace: a song in 3/4

Amazing Grace is a hymn from the 1770s, sung to an American folk tune. Ode to Joy stayed in one hand position, in 4/4, with no key signature. This song changes each of those, one at a time. It is in 3/4 time, so count "**1**, 2, 3" with a gentle lean on each 1. Here is its first line, "Amazing grace, how sweet the sound, that saved a wretch like me".

The pickup note

The song does not start on beat 1. Its first word, "A-", comes on beat 3, directly before the first full bar. A note like this, before the first bar line, is called a **pickup**. Count "1, 2" silently, play the pickup on "3", and the next note lands on the strong beat 1.

The right hand

This song is in the key of G, which has one sharp, F♯. You can see it as the key signature after the clef in Figure 7.3. The first line happens not to use F at all, so you will not need a black key yet.

Place your right thumb on D4, finger 2 on E4, finger 3 on G4, finger 4 on A4 and finger 5 on B4. (Your hand skips over F♯.) Only the last note, D5, is out of reach: lift your hand and play it with finger 5. A small move like this is normal.

bar	words	notes	fingers	beats	left hand
pickup	A-	D	1	1	–
1	-maz-ing	G, B G	3, 5 3	2, half + half	G
2	grace, how	B, A	5, 4	2, 1	G
3	sweet the	G, E	3, 2	2, 1	C
4	sound, that	D, D	1, 1	2, 1	G
5	saved a	G, B G	3, 5 3	2, half + half	G
6	wretch like	B, A	5, 4	2, 1	G
7	me	D (high)	5	3	D

"-maz-ing" has a half note on "maz", then two eighth notes, B and G, on "-ing". The last note is a dotted half note, held for the whole bar. Listen.

The left hand

The left hand plays one chord at the start of each full bar and holds it for all three beats. You need three chords, shaped so your hand barely moves:

- G chord: G2–B2–D3, fingers 5, 3, 1.
- C chord: G2–C3–E3, fingers 5, 2, 1. Keep G2, and move the other two fingers up one white key each.
- D chord: F♯2–A2–D3, fingers 5, 3, 1. Keep D3, and move the other two down: G2 down one key to F♯2, and B2 down one white key to A2.

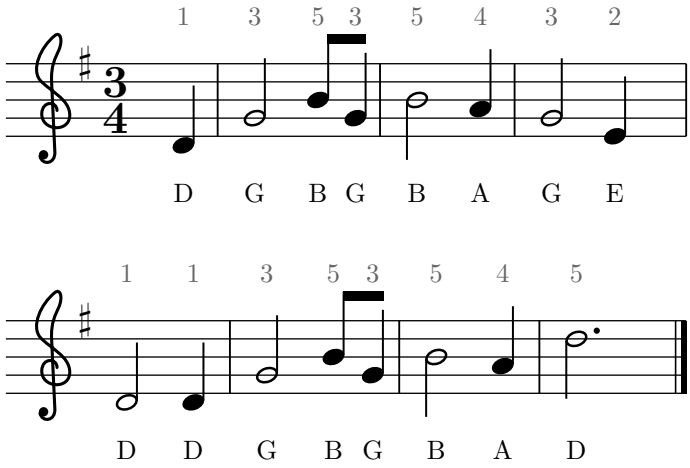


Figure 7.3: The first line of Amazing Grace in 3/4, starting with a one-beat pickup. The sharp after the clef is the key signature of G major. Finger numbers are above the staff and note names below.

Here they are in order: G, C, G, D.

The "left hand" column of the table shows which chord goes in each bar. The pickup has no chord. Listen to both hands together.

7.5 Four chords for a hundred songs

Many pop songs, old and new, use the same four chords over and over: I, V, vi and IV. In the key of C these are C, G, A minor (written Am) and F. The pieces so far had a melody to follow; this one is pure accompaniment, the part a pianist plays while someone sings. Once you can loop it smoothly, you can accompany a great many songs, and make up your own.

The right hand: smooth shapes

Choose the positions from Chapter 4 so your hand hardly moves, as in Figure 7.4:

- C: G3–C4–E4 (second inversion), fingers 1–3–5.
- G: G3–B3–D4, fingers 1–3–5. Keep your thumb on G; the other two fingers step down.
- Am: A3–C4–E4, fingers 1–3–5. All three fingers step up one white key.
- F: A3–C4–F4 (first inversion), fingers 1–2–5 as in Chapter 4. A and C stay down; only the top note moves, up from E to F.

Then back to C: the thumb steps down to G, the little finger steps down to E, and C stays.

The left hand: roots

The left hand plays the root of each chord, low down: C3 for C, G2 for G, A2 for Am, and F2 for F.

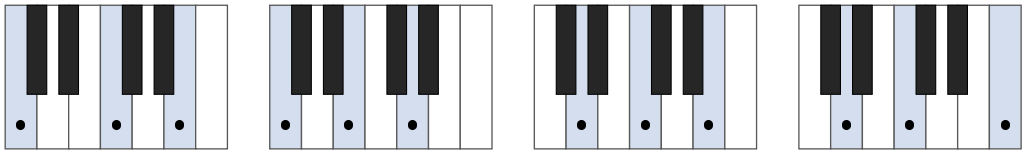


Figure 7.4: The four chords C, G, Am and F under the right hand. Neighboring chords share keys, so the hand barely moves.

Two patterns

Pattern 1 holds each chord for a whole bar. Play the left-hand root and the right-hand chord together on beat 1, and hold them while you count to 4. One chord per bar, four bars, then end on C.

Pattern 2 adds a pulse. The left hand plays its root on beat 1 and holds it. The right hand plays the chord on every beat: 1, 2, 3, 4.

Try it at the keyboard

Loop pattern 1 four times without stopping, counting "1, 2, 3, 4" in every bar. Then try pattern 2. Once it is steady, hum any melody you like over it, or improvise one with your right hand on the white keys while your left hand plays the roots.

7.6 A practice routine

Short, regular practice works far better than long, rare sessions. Fifteen to twenty minutes a day is plenty to make steady progress. Here is one way to spend them:

minutes	what	how
2	warm-up	C position, up and down, each hand, slowly
3	scales	C major and G major, hands separately (Chapter 2)
3	chords	the four-chord loop, pattern 1 then pattern 2
2	rhythm	clap and count one pattern from Chapter 5
3	reading	a few bars of new music, slowly, without stopping
5	your piece	one phrase at a time: hands alone, then together

A few habits make every minute count:

- Practice slowly enough to play it right. Speed comes from many correct, calm repetitions.
- Work on the hard bar, not the easy ones. Start your practice there, not always from the beginning.
- Use a metronome now and then, so your sense of the beat stays honest.
- Finish each session by playing something you enjoy, all the way through.

7.7 Where to go next

You have the tools to keep learning on your own. Here are some good next steps.

- *More pieces.* Many beginner pieces are free to play: the *Minuet in G* from the notebook Bach kept for his wife Anna Magdalena, the opening of Beethoven's *Für Elise*, and folk songs and hymns of every kind. Look for "easy piano" arrangements.
- *Lead sheets.* A lead sheet gives only the melody and chord symbols. Play the melody with your right hand and the chords with your left, as you did with Ode to Joy. This is how many pianists play popular songs.
- *More keys.* Use the scale recipe from Chapter 2 to play Ode to Joy and the four-chord loop in G, F and D.
- *A teacher.* Even a few lessons help a lot with posture and hand shape, which are hard to judge on your own.
- *Listen and record.* Record yourself on your phone now and then, and listen back. You will hear things you miss while playing, and you will hear your progress.

What to remember

- In the *C position*, the right thumb rests on middle C and the fingers cover C to G; the left hand mirrors it from C3 to G3.
- A *phrase* is a musical sentence. Ode to Joy has two phrases that differ only in their last bar.
- Add the left hand in stages: single bass notes first, then chords in smooth positions.
- A *pickup* is a note before the first full bar. In Amazing Grace it falls on beat 3.
- The chords I, V, vi and IV (C, G, Am, F in C) fit countless songs. Smooth positions keep the hand still.
- Practice a little every day, slowly, starting with the hardest part.

Exercises

Exercise 7.1. Move Ode to Joy to G position: right thumb on G4, then A4, B4, C5 and little finger on D5. Write the letters of the first phrase. Do the finger numbers change?

Answer

Every note moves up by the same amount, from C position to G position: C becomes G, D becomes A, E becomes B, F becomes C, and G becomes D. So the first phrase becomes:
B B C D, D C B A, G G A B, B. A A.
The finger numbers stay exactly the same (3 3 4 5, 5 4 3 2, 1 1 2 3, 3 2 2), because each finger still plays the same step of the hand position. Listen.

Exercise 7.2. In the last bar of the first phrase of Ode to Joy (E. D D), how many beats does each note get? Check that they add up to a full bar of 4/4.

Answer

The dotted quarter E gets one and a half beats, the eighth note D gets half a beat, and the half note D gets 2 beats. One and a half plus a half is 2, and 2 more makes 4: a full bar.

Exercise 7.3. Amazing Grace starts with a pickup. On which count is the pickup played, and why does the song not start on 1?

Answer

The pickup "A-" is played on count 3. In the words "a-MAZ-ing grace", the stress falls on "maz", not on "a". The strong first beat of the bar belongs on the stressed syllable, so the weak "a" comes directly before it, on the last beat of an incomplete bar.

Exercise 7.4. In the key of G, the I, V, vi and IV chords are G, D, Em and C. Find right-hand positions where neighboring chords share as many keys as possible, starting from G as D4–G4–B4.

Answer

One smooth set is:

- G: D4–G4–B4.
- D: D4–F♯4–A4. Keep D; the other two fingers step down.
- Em: E4–G4–B4. All three fingers step up.
- C: E4–G4–C5. Keep E and G; only the little finger moves, from B up to C.

Then back to G: E steps down to D, G stays, and C steps down to B. The whole loop.

Exercise 7.5. You have only ten minutes today. Which parts of the routine would you keep?

Answer

There is no single right answer. A good plan keeps a short warm-up (1 minute), spends most of the time on your piece (6 minutes, starting with the hardest bar), and ends with a few minutes of chords or reading. What matters most is playing a little every day, slowly and without tension.